HOLOMORPHIC DISKS AND THE TOP ALEXANDER GRADING OF TWO-RAINBOW $(1,1)$-KNOTS

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ABSTRACT. A two-rainbow $(1,1)$-knot is presented by a doubly pointed genus one Heegaard diagram $D[p,2,b,r]$ whose curve β meets each wall of $T^2 \setminus \alpha$ in a nested pair of rainbow arcs, and its knot Floer differential counts holomorphic disks in $\mathrm{Sym}^1(T^2) = T^2$ [23, 27]. We show that the only embedded bigons of $T^2 \setminus (\alpha \cup \beta)$ are the two lenses cut off by the inner rainbows, and that these carry the basepoints. Since $\dim_{\mathbb{C}} T^2 = 1$, an embedded index one domain admits no deformation: the nonlinear Cauchy–Riemann equation has no cokernel for any J [25, 29], every solution factors through the Riemann map,

$$\begin{tikzcd} \mathbb{R}\times[0,1] \arrow[rr, "u"] \arrow[dr, "\cong"'] & & \mathrm{Sym}^1(T^2) = T^2 \\ & \mathscr{D}(\phi) \arrow[ur, hook] & \end{tikzcd}$$

and $\#\widehat{\mathscr{M}}(\phi) = 1$ [30]. Consequently

$$\widehat{\partial} = \mathbf{0}, \qquad \mathrm{rk}\widehat{HFK}(K,s) = \#\big\{x \in \alpha\cap\beta \;\big|\; A(x) = s\big\},$$

and the four strands of one period of β carry a single integer $S = \kappa N - m$ for which

$$S \neq 0 \implies \mathrm{rk}\widehat{HFK}\big(K, A_{\mathrm{top}}\big) = 1, \qquad S = 0 \implies A(\alpha\cap\beta) = \{-1,0,+1\}, \quad g(K) = 1$$

by genus detection [24]. So $\mathrm{rk}\widehat{HFK}(K, A_{\mathrm{top}}) = 1$ or $A_{\mathrm{top}} = 1$: the rank one behaviour of an L-space knot [26, 34] survives the failure of coherence. The dichotomy is the finite Gottschalk–Hedlund alternative [6] for the Alexander cocycle over the traversal, and S decides fiberedness [31, 33].

CONTENTS

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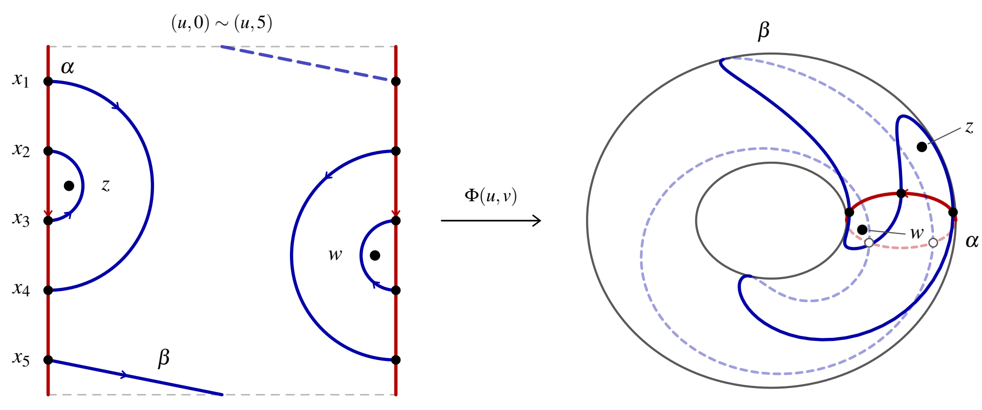
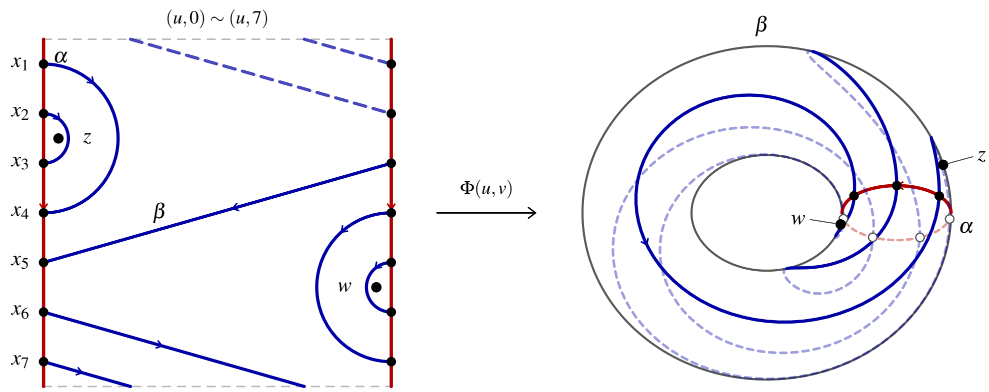
1. Introduction

1.1. **Rainbow normal forms.** A knot $K \subset S^3$ is a $(1,1)$-*knot* when it admits a doubly pointed genus-one diagram $(T^2, \alpha, \beta, z, w)$ [28, 27], and since $\mathrm{Sym}^1(T^2) = T^2$ the generators of $\widehat{CFK}(K)$ are the $p = |\alpha \cap \beta|$ intersection points [23]. Every reduced such diagram is isotopic to one with α a meridian [27, 34]. Cutting along | splits β into *rainbow arcs* and *vertical arcs* , and Doyle's normal form $[p,a,b,r]$ [35, 37] records a rainbow arcs per wall, b vertical arcs of one step size, and an offset r. Throughout $a = 2$, so each wall carries a nested pair, the inner arc cutting off the bigon that holds a basepoint and the outer one cutting off no bigon at all.



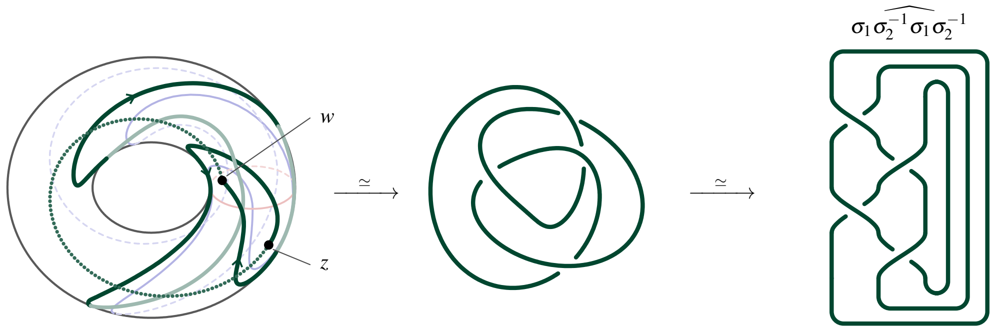
(a) $D[5,2,1,1]$ (b) $(\alpha, \beta, z, w) \subset T^2$

FIGURE 1. $[5,2,1,1]$, cut open along α and reglued by Φ. Mixed, $e = +1$, $K = J(2,2)$.



(a) $D[7,2,1,3]$ (b) $(\alpha,\beta,z,w)\subset T^2$

FIGURE 2. $[7,2,1,3]$ as in Figure 1, with two returns round the meridian. Coherent, $e=+1$.



$K\cap T^2=\{z,w\}$ $K\subset S^3$ $K=J(2,2)=4_1$

FIGURE 3. The knot of Figure 1 meets T^2 only at z and w, one arc on the surface and one dotted through the solid torus. The gradings $(1,3,1)$ give $\Delta_K=-t+3-t^{-1}$, so $K=J(2,2)$.

1.2. **Result.** Ozsváth and Szabó showed that an L-space knot has $\widehat{HFK}$ of rank one in each Alexander grading it occupies [26]. At $a=1$ every admissible tuple is coherent, so every one is an L-space knot and the rank is one throughout [42]. At $a=2$ the four rainbows need not agree, and what survives is a dichotomy.

Theorem A. Let $[p,2,b,r]$ be S^3-admissible in the sense of Definition 2.2.3 and present $K\subset S^3$, oriented so that O_L runs downwards. Then

(1) the only embedded bigons of $T^2\setminus(\alpha\cup\beta)$ are ${}_z$ and ${}_w$, so

$$\widehat{\partial}=\mathbf{0},\qquad \mathrm{rk}\widehat{HFK}(K,s)=\#\{x\mid A(x)=s\};$$

(2) the caps alternate walls along , cutting β into four strands X_0,X_1,X_2,X_3 with

$$\sigma=\begin{bmatrix}-1 & +1 & -1 & +1\end{bmatrix}^{\top},\qquad e=\sigma^{\top}\mathbf{n},\qquad \ell=\sigma^{\top}\mathbf{m};$$

(3) $=\{,\}$ or $\{,\}$ $\iff$ D is $\iff$ K is an L-space knot, and then

$$\mathrm{rk}\widehat{HFK}\big(K,A_{\mathrm{top}}\big)=1;$$

(4) $= \{\,,\,\} \implies K$ is not an L-space knot, D is of type A or B, and

$$\mathbf{n} = N\mathbf{1} - e\,\varepsilon, \quad \mathbf{m} = m\mathbf{1} - \ell\,\varepsilon, \quad \varepsilon = \begin{cases} \mathbf{e}_0, & \text{A}, \\ -\mathbf{e}_3, & \text{B}, \end{cases} \quad \begin{bmatrix} p-4 \\ b \end{bmatrix} = \begin{bmatrix} 4N \\ 4m \end{bmatrix} - \left(\mathbf{1}^\top \varepsilon\right) \begin{bmatrix} e \\ \ell \end{bmatrix};$$

(5) with $\kappa = e\,\ell$ and $S = \kappa N - m$,

$$\mathbf{m} - \kappa\mathbf{n} = -S\mathbf{1} \implies \Theta = \operatorname{diag}(\sigma)\big(\mathbf{m} - \kappa\mathbf{n}\big) = -S\,\sigma, \qquad \mathbf{1}^\top\Theta = 0 \implies \widehat{A} \circ \tau^4 = \widehat{A};$$

(6) exactly one of

$$S \neq 0 \implies \operatorname{rk}\widehat{HFK}\big(K, A_{\text{top}}\big) = 1, \qquad S = 0 \implies A(\alpha \cap \beta) = \{-1, 0, +1\}, \quad A_{\text{top}} = 1, \quad g(K) = 1;$$

(7) the second case is exactly the two ends of the band count,

$$S = 0 \iff \mathbf{1}_{\mathscr{B}} \equiv \kappa \in \{0, 1\} \iff b \in \{0,\ p-4\};$$

(8) the Dehn twist along α acts by

$$\tau_\alpha \cdot D[p, 2, b, r] = D[p, 2, b, r+1], \qquad \langle \tau_\alpha \rangle \cong \mathbb{Z}/p,$$

and (1)–(7) are invariants of K, not of the tuple.

In particular

$$\operatorname{rk}\widehat{HFK}\big(K, A_{\text{top}}\big) = 1 \qquad \text{or} \qquad A_{\text{top}} = 1.$$

Proof. §3. □

Remark 1.2.1. By Theorem A, (2) is *Hopf's Umlaufsatz*.

2. Preliminaries

We follow the Doyle–Fairchild normal form throughout, in the orientation of [42], with the wall α oriented downward and the rainbow curve β. Reversing the surface orientation and interchanging the attaching systems exchanges the two handlebodies, so the presented knot is K or its reverse $-K$; since

$$\widehat{HFK}(S^3, K, s) \cong \widehat{HFK}(S^3, -K, s)$$

[23], no statement below depends on that choice. Endpoints are one-based, as in the normal-form figures of [37], so r is the right endpoint of the first band leaving the left nest, and when $b = 0$ it is the first endpoint of the outer right rainbow. Thus for $a = 2$,

$$O_L = L_1 L_4, \qquad I_L = L_2 L_3, \qquad L_{4+i} \longleftrightarrow R_{r+i-1},$$

$$\tilde{\rho}(s) = \begin{cases} r + s - 5, & 5 \le s \le 4 + b, \\ r + s - 1, & 4 + b < s \le p, \end{cases} \qquad \rho(s) = \tilde{\rho}(s) \bmod p.$$

As a check, Figure 4 of [37] draws $D[7, 2, 1, 3]$, for which

$$L_5 \longleftrightarrow R_3, \qquad O_R = R_4 R_7, \qquad I_R = R_5 R_6.$$

Wall labels are reduced modulo p, while $v = r + b$ and $\tilde{\rho}$ denote integer lifts. Write $e = \beta \cdot \alpha \in \{\pm 1\}$, fix a path $\delta\colon z \to w$ missing α, and set

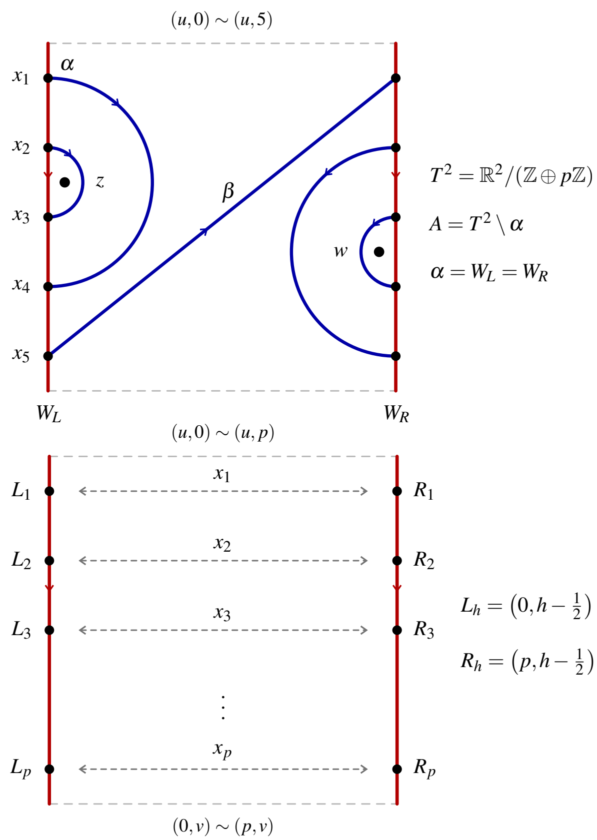
$$I_\delta(\gamma) = \gamma \cdot \delta, \qquad D_\delta = \beta \cdot \delta, \qquad \varepsilon \in \{\pm 1\}$$

for the global sign of the Alexander grading.

2.1. **Rainbow normal form.**

Convention 2.1.1 (Inherited normal form, [42, Definitions 2.1.1–2.1.4])**.** (T^2,α,β,z,w) is doubly pointed of genus one, $[p,a,b,r]$ is a rainbow tuple, and $A = T^2 \setminus \alpha = [0,1]\times\mathbb{R}/p\mathbb{Z}$ is the cut annulus with walls W_L, W_R, slots L_h, R_h at height $h-\frac{1}{2}$ increasing downwards, and identification : $R_h \mapsto L_h$. The matching has $a = 2$ and $v = r+b$,

$$L_i \leftrightarrow L_{5-i}, \quad R_{v-1+i} \leftrightarrow R_{v+4-i}, \quad i = 1,2, \qquad L_s \leftrightarrow R_{\rho(s)}, \quad 5 \le s \le p,$$
$$z \in L_2L_3, \qquad w \in R_{v+1}R_{v+2}.$$



Definition 2.1.2 (Two-rainbow arcs)**.**

$$O_L = L_1L_4 \supset I_L = L_2L_3, \qquad O_R = R_vR_{v+3} \supset I_R = R_{v+1}R_{v+2}, \qquad c_s = L_sR_{\rho(s)},$$
$$\mathscr{B} = \{c_s \mid 5 \le s \le 4+b\}, \qquad \overline{\mathscr{B}} = \{c_s \mid 4+b < s \le p\}, \qquad |\mathscr{B}| = b, \quad |\overline{\mathscr{B}}| = p-4-b,$$
$$\square_z = D(I_L) \ni z, \qquad \square_w = D(I_R) \ni w, \qquad x_z \in \partial D(I_L), \quad x_w \in \partial D(I_R),$$

and $c_s \in \overline{\mathscr{B}}$ crosses $(u,0)\sim(u,p)$ iff $\tilde{\rho}(s) > p$.

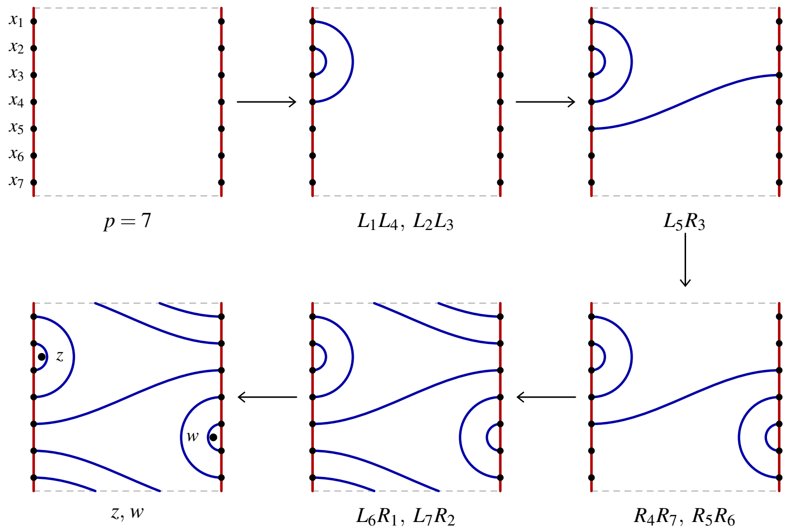


FIGURE 4. Fairchild's construction of $D[7,2,1,3]$, one arc family at a time from the marked points to the basepoints. Drawn after [37].

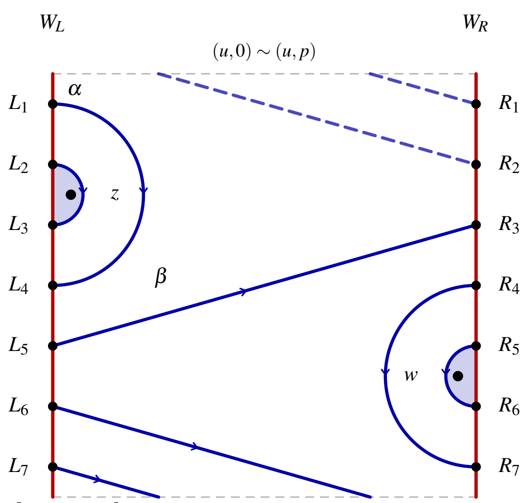


FIGURE 5. $= D[7,2,1,3]$ with $v = 4$, the nests $O_L = L_1L_4 \supset I_L = L_2L_3$ and $O_R = R_4R_7 \supset I_R = R_5R_6$ cutting off $_z$, $_w$, the band $\mathscr{B} = \{L_5R_3\}$, and the returns L_6R_1, L_7R_2 crossing $(u,0) \sim (u,p)$ with their second pieces dashed. Under the slots L_h, R_h have common image x_h.

Remark 2.1.3 (Standard position, [42, Remark 2.1.8]). α is the meridian, β carries the rainbow arcs.

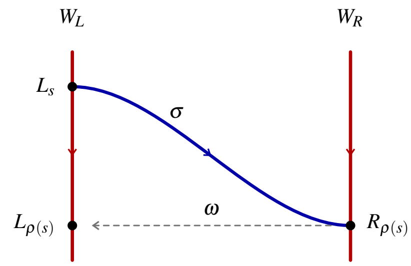
Proof. [27, §2], [34, 10]; no value of a is used. □

2.2. **Drift and admissibility.**

Definition 2.2.1 (Traversal, [42, Definition 2.2.1])**.**

$$\omega(L_h) = R_h, \qquad \omega(R_h) = L_h, \qquad \tau = \omega \circ \sigma$$

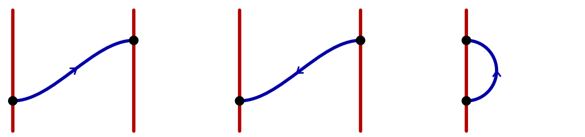
for σ the matching of Convention 2.1.1. One step is , and the *traversal* is $= \tau^{\mathbb{Z}}(L_1)$.



$\tau = \omega \circ \sigma$

Definition 2.2.2 (Drift, [42, Definition 2.2.2])**.** For $e \subset \beta \cap A$,

$$\chi(\ ) = 0, \qquad \chi(\ ) = +1, \qquad \chi(\ ) = -1, \qquad e = \sum_e \chi(e) = \beta \cdot \alpha.$$



$\chi(e) = +1$ $\chi(e) = -1$ $\chi(e) = 0$

Definition 2.2.3 (Admissibility, [42, Definition 2.2.3])**.** $D[p,2,b,r]$ is S^3-*admissible* when , that is $|\ | = p$ and $|e| = 1$.

$|\tau^{\mathbb{Z}}(L_1)| = p$ $|\tau^{\mathbb{Z}}(L_1)| < p$

Definition 2.2.4 (Band winding)**.**

$$\ell = \sum_{c_s \in \mathscr{B}} \chi(c_s) \in \mathbb{Z}, \qquad \kappa = e\ell \in \mathbb{Z}, \qquad e - \ell = \sum_{c_s \in \overline{\mathscr{B}}} \chi(c_s).$$

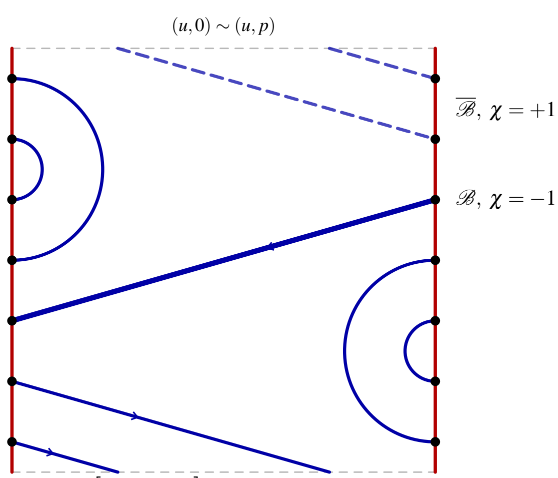


FIGURE 6. The band of $D[7,2,1,3]$ is traversed right to left and its two returns left to right, so $e = +1$, $\ell = -1$ and $\kappa = -1$.

Lemma 2.2.5 (Arc steps in the cyclic cover)**.** *Let Δh denote the height displacement of an oriented arc, read in the lift $\mathbb{R} \to \mathbb{R}/p\mathbb{Z}$. Then*

$$\Delta h(\downarrow\,\downarrow) = 0, \qquad \Delta h(c_s) = \begin{cases} \chi(c_s)\,(r-5), & c_s \in \mathscr{B}, \\ \chi(c_s)\,(r-1), & c_s \in \overline{\mathscr{B}}, \end{cases} \qquad \Delta h(c) = -\eta(c)\deg(c),$$
$$\deg(O_L) = \deg(O_R) = 3, \qquad \deg(I_L) = \deg(I_R) = 1.$$

Proof.

$$\omega(L_h) = R_h, \quad \omega(R_h) = L_h \implies \Delta h(\downarrow\,\downarrow) = h - h = 0$$
$$\Delta h(c_s) = \chi(c_s)\big(\tilde{\rho}(s) - s\big) \overset{2.1.2}{\implies} \chi(c_s)(r-5)\ (\mathscr{B}), \qquad \chi(c_s)(r-1)\ (\overline{\mathscr{B}})$$
$$O_L = L_1 L_4, \quad I_L = L_2 L_3, \quad O_R = R_v R_{v+3}, \quad I_R = R_{v+1} R_{v+2} \implies \deg = 3, 1, 3, 1$$
$$\eta(c) = +1 \overset{2.3.4}{\iff} \text{height decreases} \implies \Delta h(c) = -\eta(c)\deg(c)$$

□

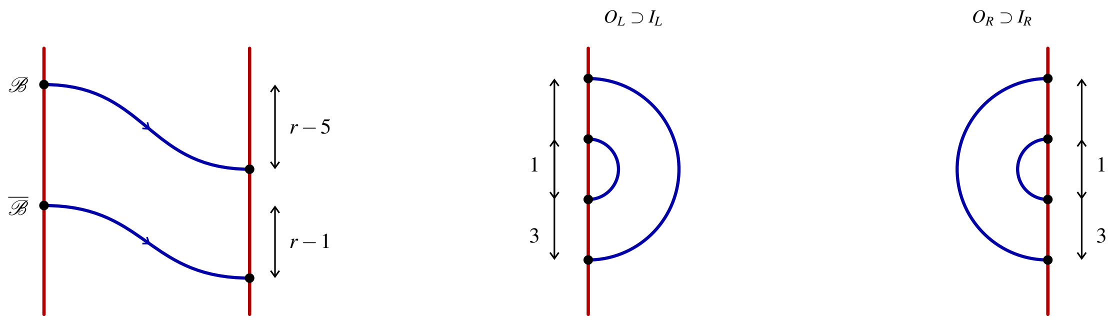


FIGURE 7. The two deck steps and the four rainbow degrees of Lemma 2.2.5.

Lemma 2.2.6 (Permutation matrix of the traversal)**.** *Order the $2p$ slots $L_1, \ldots, L_p, R_1, \ldots, R_p$, and let $T, W \in \mathbb{M}_{2p\times 2p}(\mathbb{Q})$ be the permutation matrices of τ and ω, in the sense of* [42, Convention 5.0.1],

$$T = \big(T_{mn}\big)_{m,n=1}^{2p} = \begin{bmatrix} T_{11} & T_{12} & \cdots & T_{1,2p} \\ T_{21} & T_{22} & \cdots & T_{2,2p} \\ \vdots & \vdots & \ddots & \vdots \\ T_{2p,1} & T_{2p,2} & \cdots & T_{2p,2p} \end{bmatrix}, \qquad T_{mn} = \begin{cases} 1, & n = \tau(m), \\ 0, & n \neq \tau(m). \end{cases}$$

Then

$$T \cdot T^{\top} = \mathbf{I}_{2p}, \qquad W \cdot W = \mathbf{I}_{2p}, \qquad W \cdot T \cdot W = T^{-1},$$
$$|\bigstar| = p \iff \det\big(T - \lambda \mathbf{I}_{2p}\big) = \big(\lambda^p - 1\big)^2 \iff \operatorname{rk}\big(T - \mathbf{I}_{2p}\big) = 2p - 2,$$

and then $\operatorname{spec}(T) = \{\, \zeta \in \mathbb{C} \mid \zeta^p = 1 \,\}$.

Proof.

$$\tau \text{ a bijection} \implies T \text{ has one 1 per row and column} \implies T \cdot T^{\top} = \mathbf{I}_{2p}$$
$$\omega^2 = \sigma^2 = \mathrm{id}, \quad \tau = \omega \circ \sigma \implies W \cdot W = \mathbf{I}_{2p}, \qquad W \cdot T \cdot W = \sigma \circ \omega = \tau^{-1}$$
$$\gamma \text{ a } d\text{-cycle} \implies \text{a nonzero term takes } -\lambda \text{ or 1 from every column} \implies \det\big(P - \lambda \mathbf{I}_d\big) = (-1)^d\big(\lambda^d - 1\big)$$
$$\tau = \gamma_1 \sqcup \cdots \sqcup \gamma_k, \quad \sum_{i=1}^{k} d_i = 2p \implies \det\big(T - \lambda \mathbf{I}_{2p}\big) = \prod_{i=1}^{k}\big(\lambda^{d_i} - 1\big), \qquad \dim\ker\big(T - \mathbf{I}_{2p}\big) = k$$
$$p \text{ odd} \implies \omega(\bigstar) \neq \bigstar \implies \Big[\,|\bigstar| = p \iff k = 2,\ d_1 = d_2 = p \iff \det = \big(\lambda^p - 1\big)^2 \iff \operatorname{rk} = 2p - 2\,\Big]$$

$$\lambda = 1 \text{ a root of multiplicity } k, \quad d_1 + d_2 = 2p \text{ force the converse}$$
$$\det\bigl(T - \lambda \mathbf{I}_{2p}\bigr) = 0 \iff \lambda^p = 1 \implies \operatorname{spec}(T) = \{\zeta \mid \zeta^p = 1\}$$

□

For $D[7,2,1,3]$ the traversal is $x_1x_4x_7x_2x_3x_5x_6$, so the induced permutation of $\{x_1,\dots,x_7\}$ is $1 \mapsto 4$, $2 \mapsto 3$, $3 \mapsto 5$, $4 \mapsto 7$, $5 \mapsto 6$, $6 \mapsto 1$, $7 \mapsto 2$, with

$$T = \begin{bmatrix} 0&0&0&1&0&0&0\\ 0&0&1&0&0&0&0\\ 0&0&0&0&1&0&0\\ 0&0&0&0&0&0&1\\ 0&0&0&0&0&1&0\\ 1&0&0&0&0&0&0\\ 0&1&0&0&0&0&0 \end{bmatrix}, \qquad T - \lambda \mathbf{I}_7 = \begin{bmatrix} -\lambda&0&0&1&0&0&0\\ 0&-\lambda&1&0&0&0&0\\ 0&0&-\lambda&0&1&0&0\\ 0&0&0&-\lambda&0&0&1\\ 0&0&0&0&-\lambda&1&0\\ 1&0&0&0&0&-\lambda&0\\ 0&1&0&0&0&0&-\lambda \end{bmatrix}.$$

Written out by columns, $T - \lambda \mathbf{I}_7 = \bigl[\mathbf{c}_1\ \mathbf{c}_2\ \mathbf{c}_3\ \mathbf{c}_4\ \mathbf{c}_5\ \mathbf{c}_6\ \mathbf{c}_7\bigr]$ with

$$\mathbf{c}_1 = \begin{bmatrix} -\lambda\\0\\0\\0\\0\\1\\0 \end{bmatrix}, \quad \mathbf{c}_2 = \begin{bmatrix} 0\\-\lambda\\0\\0\\0\\0\\1 \end{bmatrix}, \quad \mathbf{c}_3 = \begin{bmatrix} 0\\1\\-\lambda\\0\\0\\0\\0 \end{bmatrix}, \quad \mathbf{c}_4 = \begin{bmatrix} 1\\0\\0\\-\lambda\\0\\0\\0 \end{bmatrix}, \quad \mathbf{c}_5 = \begin{bmatrix} 0\\0\\1\\0\\-\lambda\\0\\0 \end{bmatrix}, \quad \mathbf{c}_6 = \begin{bmatrix} 0\\0\\0\\0\\1\\-\lambda\\0 \end{bmatrix}, \quad \mathbf{c}_7 = \begin{bmatrix} 0\\0\\0\\1\\0\\0\\-\lambda \end{bmatrix},$$

each column carrying $-\lambda$ in its own row and 1 in row $\tau^{-1}(j)$, where τ^{-1} is $1 \mapsto 6$, $2 \mapsto 7$, $3 \mapsto 2$, $4 \mapsto 1$, $5 \mapsto 3$, $6 \mapsto 5$, $7 \mapsto 4$. A nonzero Leibniz term takes one entry from each column, so it takes $-\lambda$ from every column or 1 from every column, and

$$\prod_{j=1}^{7}(-\lambda) = (-\lambda)^7 = -\lambda^7, \qquad \operatorname{sgn}(\mathrm{id}) = +1, \qquad \prod_{j=1}^{7} 1 = 1, \qquad \operatorname{sgn}\bigl(\tau^{-1}\bigr) = \operatorname{sgn}(1\,4\,7\,2\,3\,5\,6) = (-1)^{7-1} = +1,$$

$$\det\bigl(T - \lambda \mathbf{I}_7\bigr) = (+1)\cdot\bigl(-\lambda^7\bigr) + (+1)\cdot 1 = -\lambda^7 + 1 = (-1)^7\bigl(\lambda^7 - 1\bigr).$$

The kernel of $T - \mathbf{I}_7$ is cut out by $x_4 = x_1$, $x_3 = x_2$, $x_5 = x_3$, $x_7 = x_4$, $x_6 = x_5$, $x_1 = x_6$, $x_2 = x_7$, which chain to $x_1 = x_4 = x_7 = x_2 = x_3 = x_5 = x_6$, so it is spanned by $\mathbf{1} = (1,1,1,1,1,1,1)^\top$ and

$$\bigl(T - \mathbf{I}_7\bigr)\cdot \mathbf{1} = \begin{bmatrix} -1&0&0&1&0&0&0\\ 0&-1&1&0&0&0&0\\ 0&0&-1&0&1&0&0\\ 0&0&0&-1&0&0&1\\ 0&0&0&0&-1&1&0\\ 1&0&0&0&0&-1&0\\ 0&1&0&0&0&0&-1 \end{bmatrix} \cdot \begin{bmatrix} 1\\1\\1\\1\\1\\1\\1 \end{bmatrix} = \begin{bmatrix} 0\\0\\0\\0\\0\\0\\0 \end{bmatrix} = \mathbf{0}, \qquad \operatorname{rk}\bigl(T - \mathbf{I}_7\bigr) = 7 - 1 = 6.$$

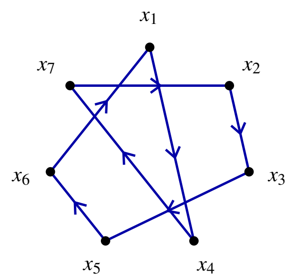


FIGURE 8. The traversal of $D[7,2,1,3]$ as the 7-cycle $(1\,4\,7\,2\,3\,5\,6)$.

Lemma 2.2.7 (Rank, [42, Lemma 2.2.4])**.**

$$\implies \big|\pi_0(\beta)\big| = 1, \qquad (T^2,\alpha,\beta) \simeq S^3, \qquad \mathrm{rk}\big(\widehat{CFK}(K)\big) = p.$$

Proof.

$$\overset{2.2.6}{\implies} \tau = \gamma_1 \sqcup \gamma_2, \quad |\gamma_1| = |\gamma_2| = p \implies \big|\pi_0(\beta)\big| = 1, \qquad \alpha \cap \beta = \{x_1,\dots,x_p\} \implies \mathrm{rk}\big(\widehat{CFK}(K)\big) = p$$

$$|e| = |\beta\cdot\alpha| = 1 \implies H_1(T^2) = \mathbb{Z}\langle\alpha\rangle \oplus \mathbb{Z}\langle\beta\rangle \implies (T^2,\alpha,\beta) \simeq S^3$$

□

Lemma 2.2.8 (Holonomy of the lifted traversal)**.**

$$e(r-1) - 4\ell - 3\big(\eta(O_L)+\eta(O_R)\big) - \big(\eta(I_L)+\eta(I_R)\big) \equiv 0 \pmod p,$$

$$= \{\ ,\ \}, \quad = \{\ ,\ \} \iff e(r-1) - 4\ell + 8 \equiv 0 \pmod p,$$

$$O_L = \ , \quad I_L = \ , \quad O_R = \ , \quad I_R = \ \iff e(r-1) - 4\ell + 4 \equiv 0 \pmod p.$$

Proof.

$$\text{closed}, \quad \Delta h(\ ) = 0 \overset{2.2.5}{\implies} 0 \equiv \sum_{s=5}^{p} \Delta h(c_s) + \sum_c \Delta h(c) \pmod p$$

$$\sum_{s=5}^{p} \Delta h(c_s) = (r-5)\,\ell + (r-1)(e-\ell) = e(r-1) - 4\ell, \qquad \sum_c \Delta h(c) = -3\big(\eta(O_L)+\eta(O_R)\big) - \big(\eta(I_L)+\eta(I_R)\big)$$

$$\overset{2.3.4}{\implies} O_L, I_L, O_R, I_R \text{ all } \ ,\ \implies -3(-2) - (-2) = +8$$

$$O_L = \ , \quad O_R = \ , \quad I_L = \ , \quad I_R = \ \implies -3(-2) - (+2) = +4$$

$$\overset{2.3.5}{\implies} \text{the balanced patterns give } 8, 4, 0, -4, -8$$

$$p \text{ odd}, \quad p \geq 5 \implies 8, 4, 0, -4, -8 \text{ pairwise distinct} \pmod p \implies \text{each congruence determines its pattern}$$

□

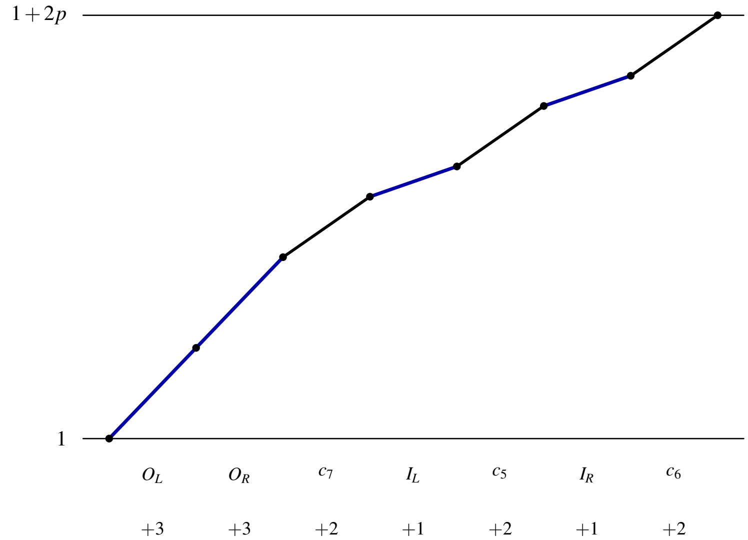


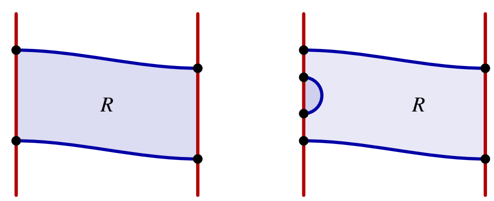
FIGURE 9. The lifted traversal of $D[7,2,1,3]$ rises by $e(r-1)-4\ell+8 = 2+4+8 = 14 = 2p$ and so closes modulo p.

2.3. **Reduced and coherent diagrams.**

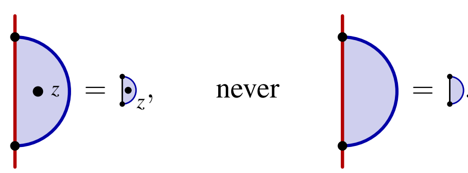
Definition 2.3.1 (Region type, [42, Definition 2.3.1])**.** A component R of $T^2 \setminus (\alpha \cup \beta)$ is *n-sided* when $|\partial R \cap \alpha \cap \beta| = 2n$,

$$(n=1), \qquad (n=2), \qquad (n=3), \qquad (n=4),$$

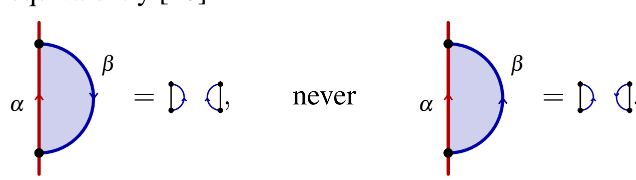
the darker lobe against a wall being a neighbouring bigon, one of the n sides of R and not part of it.



Definition 2.3.2 (Reduced diagram, [34])**.** A diagram is *reduced* when every bigon of $T^2 \setminus (\alpha \cup \beta)$ carries a basepoint,



Definition 2.3.3 (Coherent diagram, [34])**.** A reduced diagram is *coherent* when α, β admit orientations coorienting every bigon, equivalently [10]



Definition 2.3.4 (Rainbow sense)**.**

$$= \{O_L, I_L\}, \qquad = \{O_R, I_R\}, \qquad \eta() = \eta() = +1, \qquad \eta() = \eta() = -1,$$
$$u_L = \#\{c \in \;|\; c = \}, \qquad d_L = \#\{c \in \;|\; c = \} = 2 - u_L,$$
$$u_R = \#\{c \in \;|\; c = \}, \qquad d_R = \#\{c \in \;|\; c = \} = 2 - u_R.$$

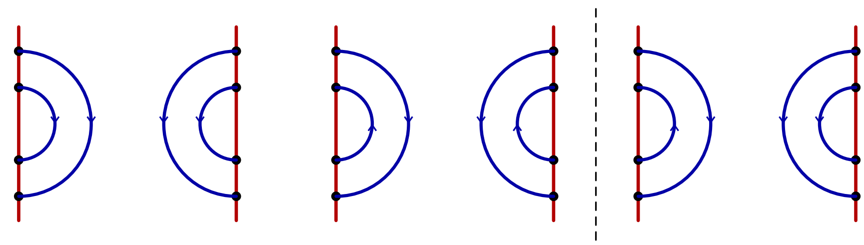


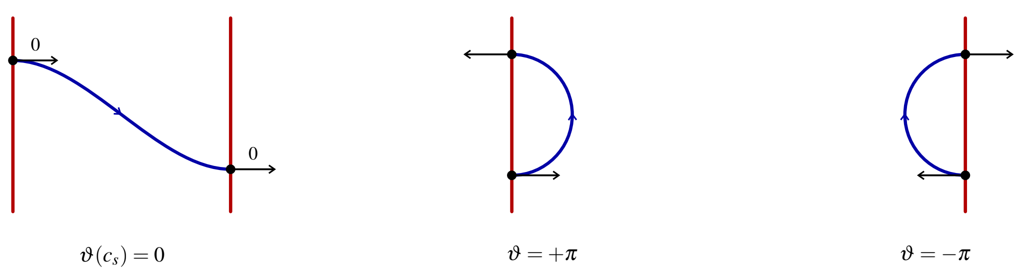
FIGURE 10. Two balanced sense patterns and, to the right of the rule, the pattern excluded by Lemma 2.3.5.

Lemma 2.3.5 (Turning balance)**.**

$$\vartheta(\beta) = 2\pi\big(u_L - u_R\big) = 0 \iff u_L = u_R \in \{0,1,2\},$$
$$= \{,\} \iff = \{,\}, \qquad = \{,\} \iff = \{,\},$$
$$= \{,\} \iff = \{,\}.$$

Proof.

β embedded essential $\implies \beta \simeq$ geodesic $\implies \vartheta(\beta) = 0, \qquad \vartheta() = 0, \quad \vartheta() = +\pi\eta, \quad \vartheta() = -\pi\eta$



$0 = \vartheta(\beta) = \pi\big(u_L - d_L\big) - \pi\big(u_R - d_R\big) = 2\pi\big(u_L - u_R\big) \implies u_L = u_R \in \{0,1,2\}$ [42, Thm. A.3.3, Lem. 3.2.3]

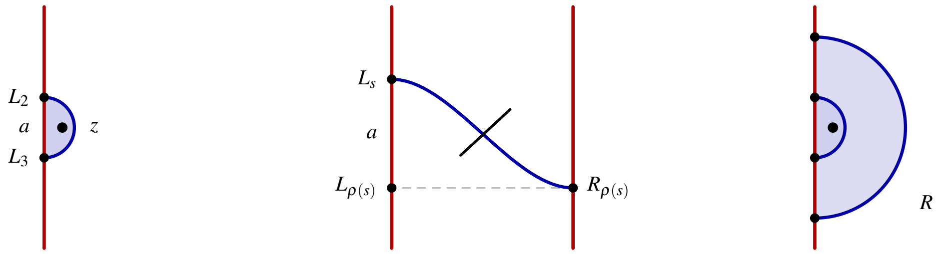
□

Theorem 2.3.6 (Bigon classification and reducedness)**.**

$$\textit{embedded} \implies \in \{{}_z, {}_w\}, \qquad , \qquad \widehat{\partial} = \mathbf{0}.$$

Proof.

$\partial = a \cup e, \quad \mathrm{int}(e) \cap \alpha = \varnothing \overset{2.1.2}{\implies} e \in \cup \cup \{\}, \qquad (a \cup c_s) \cdot \mu = \pm 1 \implies e \neq c_s$ [42, Lem. 3.1.1]

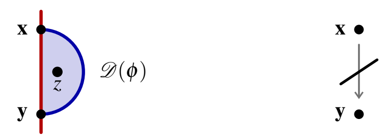


$[a \cup I_L] = 0$ $[a \cup c_s] \neq 0$ $D_z \subset D(O_L)$

$$D(O_L) \setminus D(I_L) = D(O_R) \setminus D(I_R) = \implies e \in \{I_L, I_R\} \implies \in \{ _z, _w\} \overset{2.3.2}{\iff}$$

$$\mu(\varphi) = 1, \quad \mathrm{Sym}^1(T^2) = T^2 \implies \mathscr{D}(\varphi) = \implies n_z(\varphi) + n_w(\varphi) > 0 \implies \widehat{\partial} = \mathbf{0} \qquad [28, 34]$$

□



$\mu(\phi) = 1, n_z(\phi) = 1$ $\widehat{\partial}\mathbf{x} = 0$

Corollary 2.3.7 (Rank equals count)**.**

$$\mathrm{rk}\widehat{HFK}(K,s) = \#\{x \in \alpha \cap \beta \mid A(x) = s\}, \qquad \sum_s \mathrm{rk}\widehat{HFK}(K,s) = p.$$

Proof.

$$\widehat{\partial} = \mathbf{0} \overset{2.3.6}{\implies} \widehat{HFK}(K) = \widehat{CFK}(K) = \bigoplus_{k=1}^{p} \mathbb{F} \cdot x_k \overset{2.2.7}{\implies} \mathrm{rk}\widehat{HFK}(K,s) = \#\{x \mid A(x) = s\} \qquad [42, \text{Cor. } 3.1.3]$$

□

Proposition 2.3.8 (Coherence)**.**

$$= \{ , \}, = \{ , \} \quad or \quad = \{ , \}, = \{ , \}$$

$$\iff \eta(O_L) = \eta(I_L) = \eta(O_R) = \eta(I_R) = \sigma \iff , \; never \implies \; coherent.$$

Proof.

$$\overset{2.3.6}{\implies} \in \{ _z, _w\}, \qquad a(D) = \langle \beta, \partial^+ D \rangle, \quad b(D) = \langle \alpha, \partial^+ D \rangle, \qquad : a = \sigma, \, b = -\tau, \quad : a = -\sigma, \, b = \tau$$

$a = \sigma, \, b = -\tau$ $a = -\sigma, \, b = \tau$

$$\partial _z, \; \partial _w \text{ cooriented} \iff \sigma\tau = -1 \iff \tau := -\sigma \implies a(D) = b(D) \;\; \forall D \overset{2.3.3}{\iff}$$

$$\sigma \text{ exists} \overset{2.3.4}{\iff} \in \{\{ , \}, \{ , \}\} \overset{2.3.5}{\iff} \in \{\{ , \}, \{ , \}\}$$

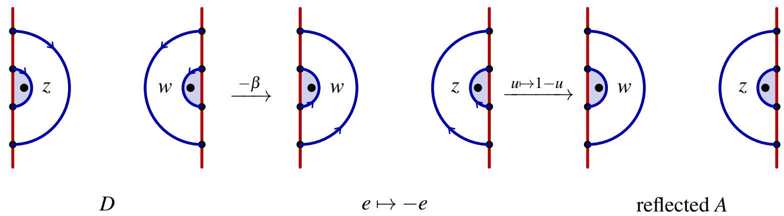
□

2.4. **Symmetries.**

Convention 2.4.1 (Reversal and mirror, [42, Definition 2.4.1])**.**

$$-\beta: \quad \longmapsto, \qquad \leftarrow, \qquad \longmapsto, \qquad \longmapsto,$$

and reflecting A by $u \mapsto 1-u$ is .



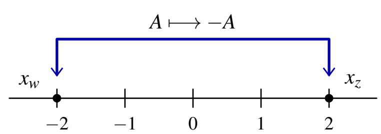
Lemma 2.4.2 (Effect on the gradings, [42, Lemma 2.4.2])**.**

$$-\beta: \qquad e \longmapsto -e, \qquad A \longmapsto -A, \qquad x_z \longleftrightarrow x_w.$$

Proof.

$$\overset{2.2.2}{\Longrightarrow} e = \sum_{s=5}^{p} \chi(c_s) \longmapsto -e, \qquad \leftarrow \implies (n_z, n_w) \longmapsto (n_w, n_z) \implies A \longmapsto -A \overset{2.1.1}{\Longrightarrow} x_z \longleftrightarrow x_w$$

□



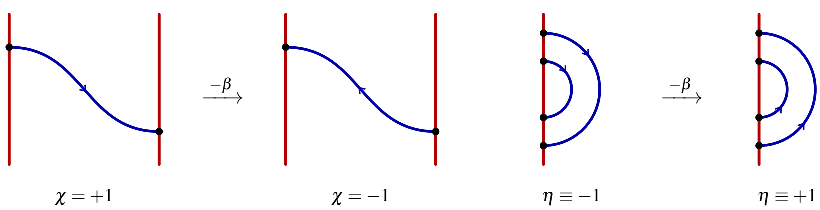
Lemma 2.4.3 (Reversal invariants)**.**

$$-\beta: \qquad \chi \longmapsto -\chi, \qquad e \longmapsto -e, \qquad \ell \longmapsto -\ell, \qquad \kappa \longmapsto \kappa, \qquad \eta \longmapsto -\eta,$$

$$= \{,\} \iff -\beta: = \{,\}, \qquad = \{,\} \iff -\beta: = \{,\}.$$

Proof.

$$\overset{2.2.2,\,2.2.4}{\Longrightarrow} \chi \longmapsto -\chi, \quad \mathscr{B} \text{ setwise fixed} \implies e \longmapsto -e, \quad \ell \longmapsto -\ell, \quad \kappa = e\,\ell \longmapsto (-e)(-\ell) = \kappa$$



$$\overset{2.4.1,\,2.3.4}{\Longrightarrow} \longleftrightarrow, \qquad \longleftrightarrow \implies \eta \longmapsto -\eta$$

$$= \{,\} \longmapsto \{,\}, \qquad = \{,\} \longmapsto \{,\} = \{,\}$$

□

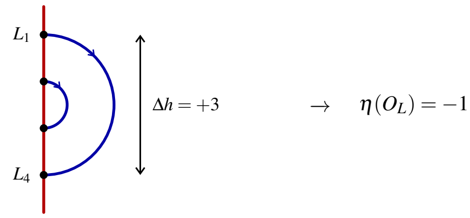
Corollary 2.4.4 (Reversal normalisation)**.**

$$= \{,\} \iff -\beta: = \{,\} \implies e = +1, \qquad \kappa = \ell, \qquad O_L = .$$

Proof.

$$= \{,\} \overset{2.4.3}{\Longrightarrow} \text{ fixed}, \qquad = \{,\} \overset{2.4.3}{\Longrightarrow} = \{,\} \overset{2.3.8}{\Longrightarrow}$$

$$\overset{2.4.2}{\Longrightarrow} -\beta \text{ presents } -K, \qquad \widehat{HFK}(S^3, -K, s) \cong \widehat{HFK}(S^3, K, s) \implies e = +1 \overset{2.2.4}{\Longrightarrow} \kappa = e\,\ell = \ell$$



□

Remark 2.4.5 (The drift is not determined). Reversal fixes the mixed sense pattern (Lemma 2.4.3), so nothing intrinsic picks a sign of e: Corollary 2.4.4 is a normalisation, not a theorem.

2.5. **Knot Floer input.**

Theorem 2.5.1 (Coherence criterion, [34, Theorem 1.2], [42, Theorem 2.5.1])**.**

$$D\ , \quad D\ \ \iff K \text{ an } L\text{-space knot}.$$

Proof. [34, Theorem 1.2]. □

Theorem 2.5.2 (Alexander polynomial, [26, Theorem 1.2], [42, Theorem 2.5.2])**.**

$$K \text{ an } L\text{-space knot} \implies \Delta_K(t) = \sum_{j=0}^{2n} (-1)^j t^{A_j}, \qquad A_0 > A_1 > \cdots > A_{2n},$$

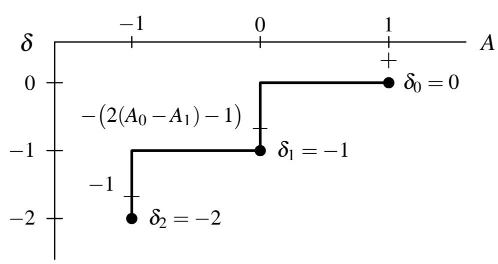
$$\operatorname{rk}\widehat{HFK}(K, A_j) = 1, \qquad \widehat{HFK}(K, s) = 0, \quad s \notin \{A_0, \dots, A_{2n}\}.$$

Proof. [26, Theorem 1.2]. □

Definition 2.5.3 (Alexander criterion, [42, Definition 2.5.3])**.** Δ_K has the form of Theorem 2.5.2, that is .

Definition 2.5.4 (Maslov normalisation, [42, Definition 2.5.4])**.**

$$A = A_{\text{top}} \implies M = 0, \qquad \delta_0 = 0, \qquad \delta_j = \begin{cases} \delta_{j-1} - 2\big(A_{j-1} - A_j\big) + 1, & j \text{ odd}, \\ \delta_{j-1} - 1, & j \text{ even}. \end{cases}$$



$K = T(2,3)$, $\Delta_K(t) = t - 1 + t^{-1}$

Remark 2.5.5 (Mirrors, [42, Remark 2.5.5]).

$$: \quad \big(\delta_0, \delta_1, \dots, \delta_{2n}\big) \longmapsto \big(\delta_{2n}, \dots, \delta_1, \delta_0\big),$$

so Definition 2.5.4 precedes Definition 2.5.3.

Proof. Lemma 2.4.2. □

Lemma 2.5.6 (Balanced chains determine domains)**.**

$$c \subset \beta, \quad \partial c = y - x, \quad c \cdot \alpha = 0 \implies \exists \mathscr{D} \in \pi_2(x, y), \quad \partial_\beta \mathscr{D} = c, \quad \mathscr{D} \text{ unique modulo } \mathbb{Z}[T^2].$$

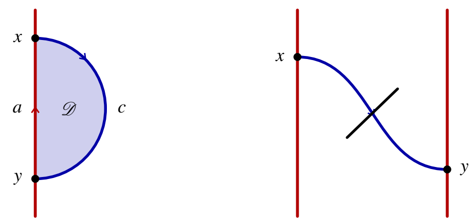
Proof.

$$\partial(c+a_0) = (y-x)+(x-y) = 0 \implies [c+a_0] = m\,[\alpha]+n\,[\beta], \qquad 0 = c\cdot\alpha + a_0\cdot\alpha = [c+a_0]\cdot[\alpha] = n\,e \implies n=0$$

$$a := a_0 - m\,\alpha \implies \partial a = x-y, \quad [c+a]=0 \implies c+a = \partial\mathscr{D}$$

$$\mathrm{Sym}^1(T^2) = T^2 \implies \mathscr{D} \longleftrightarrow \varphi_{\mathscr{D}} \in \pi_2(x,y), \qquad \partial_\beta\mathscr{D} = c \qquad \text{[28, §2], [23, §3]}$$

$$\partial\mathscr{D} = \partial\mathscr{D}' \implies \partial(\mathscr{D}-\mathscr{D}') = 0 \implies \mathscr{D}-\mathscr{D}' \in H_2(T^2;\mathbb{Z}) = \mathbb{Z}[T^2]$$



$\partial\mathscr{D} = c + a$ $\qquad$ $c\cdot\alpha = \chi(c) = +1$

□

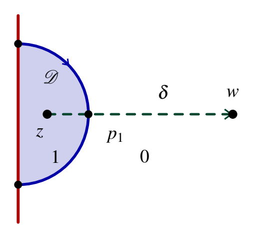
Lemma 2.5.7 (Path-pairing formula)**.**

$$n_z(\mathscr{D}) - n_w(\mathscr{D}) = \varepsilon_0\left(\partial_\beta\mathscr{D}\right)\cdot\delta, \qquad \varepsilon_0 \in \{\pm 1\} \textit{ independent of } \mathscr{D}.$$

Proof.

$$n_\delta(t) = \text{multiplicity of } \mathscr{D} \text{ at } \delta(t), \qquad \delta\cap\alpha = \varnothing \implies n_\delta \text{ jumps only at } \delta \pitchfork \partial_\beta\mathscr{D}, \text{ by } \varepsilon_0\, s_i$$

$$n_z(\mathscr{D}) - n_w(\mathscr{D}) = n_\delta(0) - n_\delta(1) = \varepsilon_0\sum_{i=1}^{k} s_i = \varepsilon_0\left(\partial_\beta\mathscr{D}\right)\cdot\delta$$



$n_z(\mathscr{D}) - n_w(\mathscr{D}) = n_\delta(0) - n_\delta(1) = 1 - 0 = \varepsilon_0\, s_1$

□

Proposition 2.5.8 (Connection formula)**.**

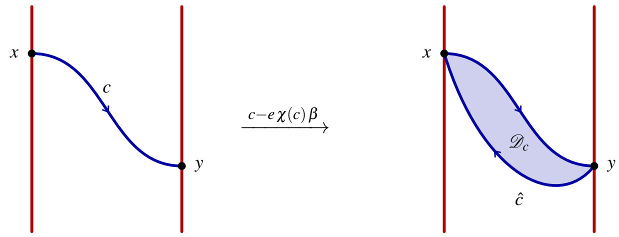
$$c \subset \beta, \quad c\colon x \to y, \quad \chi(c) = c\cdot\alpha \implies A(y) - A(x) = \varepsilon\big(I_\delta(c) - e\,D_\delta\,\chi(c)\big).$$

Proof.

$$\hat{c} := c - e\,\chi(c)\,\beta \implies \hat{c}\cdot\alpha = \chi(c)\big(1-e^2\big) = 0, \quad \partial\hat{c} = y - x \overset{2.5.6}{\implies} \hat{c} = \partial_\beta\mathscr{D}_c, \quad \mathscr{D}_c \in \pi_2(x,y)$$

$$\mathscr{D}_c + m\,[T^2] \implies n_z - n_w \text{ unchanged} \overset{2.5.7}{\implies} A(y) - A(x) = \varepsilon\,\hat{c}\cdot\delta, \quad \varepsilon \in \{\pm 1\} \text{ independent of } c \quad \text{[23, §3]}$$

$$\hat{c}\cdot\delta = c\cdot\delta - e\,\chi(c)\,(\beta\cdot\delta) = I_\delta(c) - e\,D_\delta\,\chi(c)$$



$c \cdot \alpha = \chi(c) = +1$ $\qquad$ $\hat{c} \cdot \alpha = \chi(c)(1-e^2) = 0$

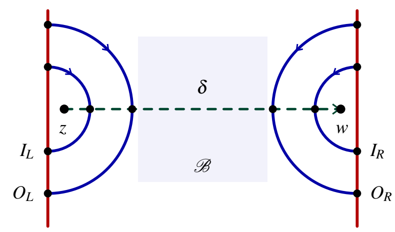
□

Lemma 2.5.9 (Rainbow step)**.**
$c \in \{O_L, I_L, O_R, I_R\}, \qquad \varepsilon := +1 \implies \Theta(c) = \eta(c), \qquad \Theta() = \Theta() = +1, \qquad \Theta() = \Theta() = -1.$

Proof.

$$\chi(c) = 0 \overset{2.5.8}{\implies} \Theta(c) = \varepsilon I_\delta(c), \qquad {}_z \subset D(I_L) \subset D(O_L), \qquad {}_w \subset D(I_R) \subset D(O_R) \implies \#(\delta \cap c) = 1$$

$$\operatorname{sgn}(c'(p) \wedge \delta'(p)) = \eta(c) \overset{2.3.4}{\implies} I_\delta(c) = \eta(c), \qquad \varepsilon := +1 \implies \Theta(c) = \eta(c)$$



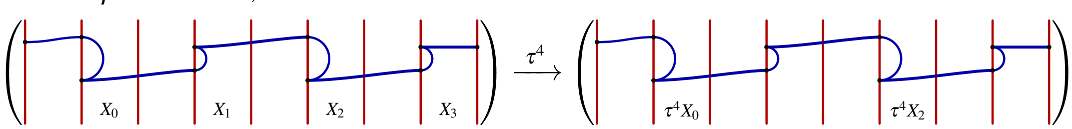
$\#(\delta \cap c) = 1, \quad I_\delta(c) = \eta(c), \quad c \in \{O_L, I_L, O_R, I_R\}$

□

2.6. **Combinatorial input.** From here $D[p,2,b,r]$ is mixed and normalised as in Corollary 2.4.4,

$$= \{ , \}, \qquad = \{ , \}, \qquad O_L = , \qquad e = +1, \qquad \kappa = e\ell = \ell.$$

A maximal run of between two consecutive , along is a *strand*. One period of carries exactly four of them; index them by $i \in \mathbb{Z}/4$ in the cyclic order in which meets them, so that τ^4 carries one period of $\widehat{\beta}$ to the next,

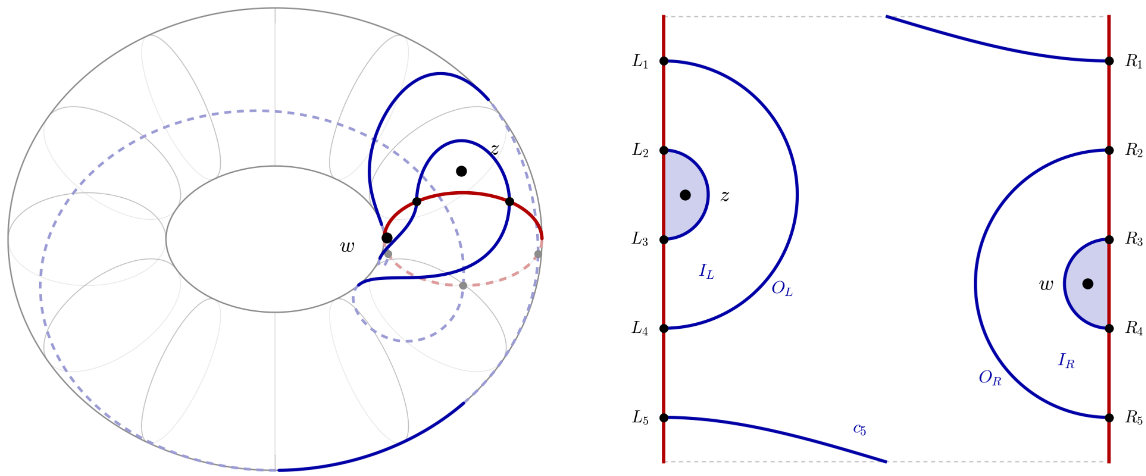


Write n_i for the number of in X_i and $m_i = \sum_{c \in X_i} \mathbf{1}_{\mathscr{B}}(c)$ for the number of those that are bands, and collect the four values of each into a vector; the sign vector $\boldsymbol{\sigma}$ records the constant corner sign of each strand,

$$\mathbf{n} = \begin{bmatrix} n_0 \\ n_1 \\ n_2 \\ n_3 \end{bmatrix} \in \mathbb{Z}^4, \qquad \mathbf{m} = \begin{bmatrix} m_0 \\ m_1 \\ m_2 \\ m_3 \end{bmatrix} \in \mathbb{Z}^4, \qquad \boldsymbol{\sigma} = \begin{bmatrix} \chi_{X_0} \\ \chi_{X_1} \\ \chi_{X_2} \\ \chi_{X_3} \end{bmatrix} = \begin{bmatrix} -1 \\ +1 \\ -1 \\ +1 \end{bmatrix}, \qquad \mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

Every scalar below is then one linear functional of this pair,

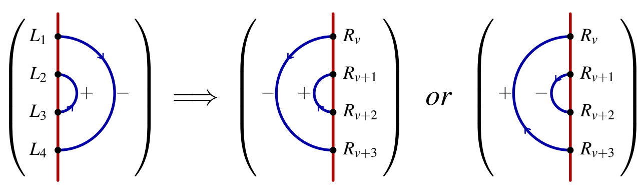
$$e = \boldsymbol{\sigma}^\top \mathbf{n}, \qquad \ell = \boldsymbol{\sigma}^\top \mathbf{m}, \qquad p = \mathbf{1}^\top \mathbf{n} + 4, \qquad b = \mathbf{1}^\top \mathbf{m}, \qquad \kappa = e\ell.$$



(a) $(T^2, \alpha, \beta, z, w)$, $D[5,2,1,1]$ (b) $A = T^2 \smallsetminus \alpha$, $R_h \mapsto L_h$

FIGURE 11. The nests hang off the same on opposite sides, so $_z \subset D(I_L) \subset D(O_L)$ and $_w \subset D(I_R) \subset D(O_R)$.

Lemma 2.6.1 (Corner alternation along $\partial\beta$)**.**



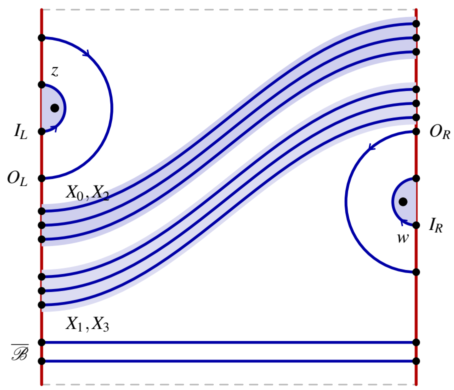
in this cyclic order along $,\qquad \sigma = \begin{bmatrix} -1 & +1 & -1 & +1 \end{bmatrix}^\top.$

Proof.

$$O_L = \overset{2.3.4}{\Longrightarrow} \big(\eta(O_L), \eta(I_L)\big) = (-1,+1), \qquad = \{\,,\,\} \overset{2.3.5}{\Longrightarrow} \big(\eta(O_R), \eta(I_R)\big) \in \{(-1,+1),(+1,-1)\}$$

$$\text{ends on } W_L \longrightarrow W_R, \quad \chi = -1 \text{ until } , \qquad \text{ends on } W_R \longrightarrow W_L, \quad \chi = +1 \text{ until }$$

$$\Longrightarrow \;,\,,\,, \text{ alternate along } , \qquad \sigma = \begin{bmatrix} -1 & +1 & -1 & +1 \end{bmatrix}^\top$$

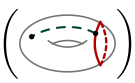
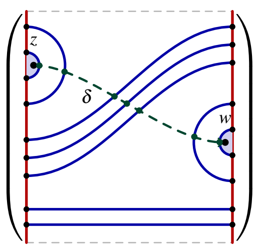


$\sigma_i = (-1)^{i+1}$

□

Lemma 2.6.2 (Alexander connecting arc)**.**

$$\mathscr{B} = \{c_s \mid 5 \le s \le 4+b\}, \qquad \overline{\mathscr{B}} = \{c_s \mid 5+b \le s \le p\}, \qquad \tilde{\rho}(\mathscr{B}) = \{r, r+1, \ldots, r+b-1\},$$

$\exists\, \delta \subset \Big($  $\Big),\qquad \delta = \Bigg($  $\Bigg)$: *$z \to w$ embedded*, $\qquad \delta \cap \alpha = \varnothing,$

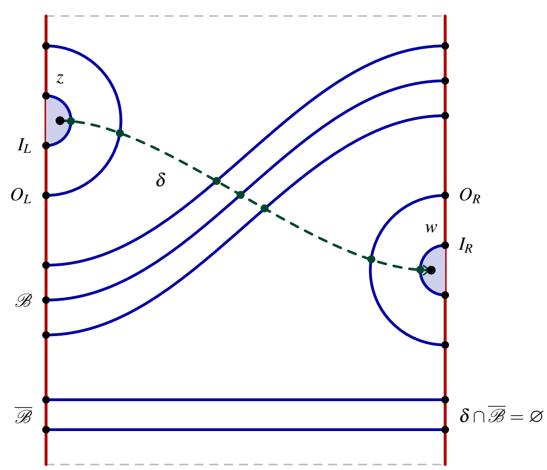
$$\delta \pitchfork \beta = \{I_L, O_L\} \cup \mathscr{B} \cup \{O_R, I_R\}, \qquad \#\big(\delta \cap c_s\big) = \mathbf{1}_{\mathscr{B}}(c_s), \qquad 5 \le s \le p.$$

Proof.

$$\tilde{\rho}(s) = r+s-5\ (\mathscr{B}), \quad \tilde{\rho}(s) = r+s-1\ (\overline{\mathscr{B}}) \implies \tilde{\rho} \text{ strictly increasing} \implies A \setminus \bigcup_{s=5}^{p} c_s = \bigsqcup_{j=1}^{p-4} F_j$$

$$z \in \mathbf{D}_z \subset F_L, \qquad w \in \mathbf{D}_w \subset F_R, \qquad \tilde{\rho}(c_{4+b}) = v-1 \implies \operatorname{sep}(F_L, F_R) = \mathscr{B}$$

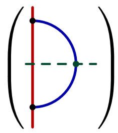
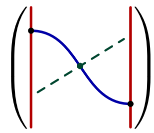
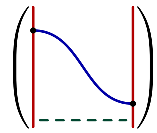
$$\delta = \eta_z * \gamma^{\vee} * \eta_w \text{ embedded}, \qquad \delta \cap \alpha = \varnothing, \qquad \delta \pitchfork \beta = \{I_L, O_L\} \cup \mathscr{B} \cup \{O_R, I_R\}$$



$\#(\delta \cap c_s) = \mathbf{1}_{\mathscr{B}}(c_s)$

□

Corollary 2.6.3 (Alexander increments of the elementary domains)**.** *Each bracket is* Θ *of the arc it draws,*

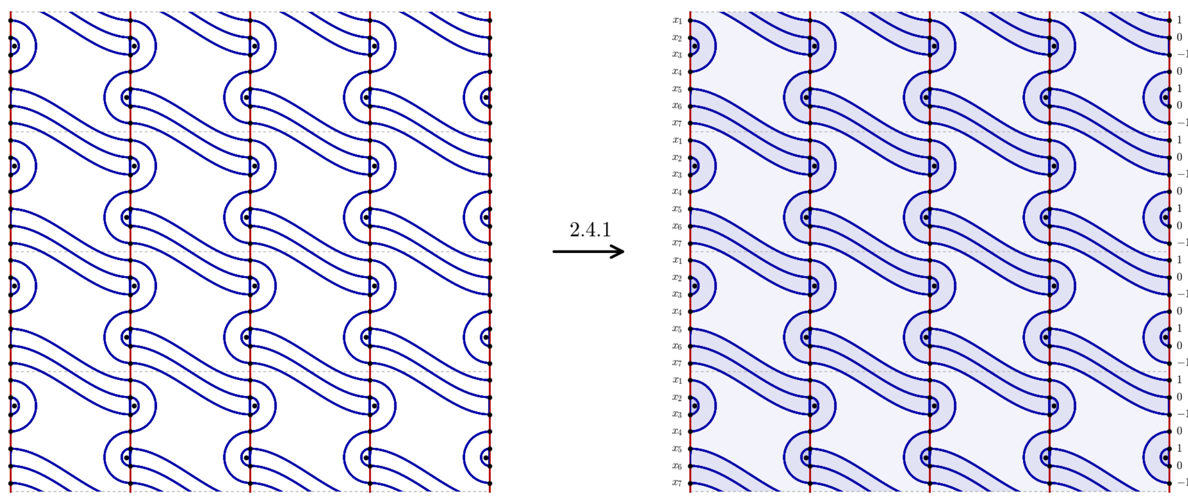
 $= \eta(c)$,  $= \chi(c_s)\big(1-\kappa\big)$,  $= -\chi(c_s)\,\kappa.$

Proof.

$$\xRightarrow{2.6.2,\,2.5.9} D_\delta = \beta\cdot\delta = \underbrace{(-1)+(+1)+(-1)+(+1)}_{\eta \text{ of the four caps}} + \sum_{c_s\in\mathscr{B}} \chi(c_s) = \ell$$

$$\xRightarrow{2.5.8} \Theta(c_s) = \chi(c_s)\,\mathbf{1}_{\mathscr{B}}(c_s) - e\,\ell\,\chi(c_s) = \chi(c_s)\big(\mathbf{1}_{\mathscr{B}}(c_s)-\kappa\big), \qquad \chi(c) = 0 \implies \Theta(c) = \eta(c)$$

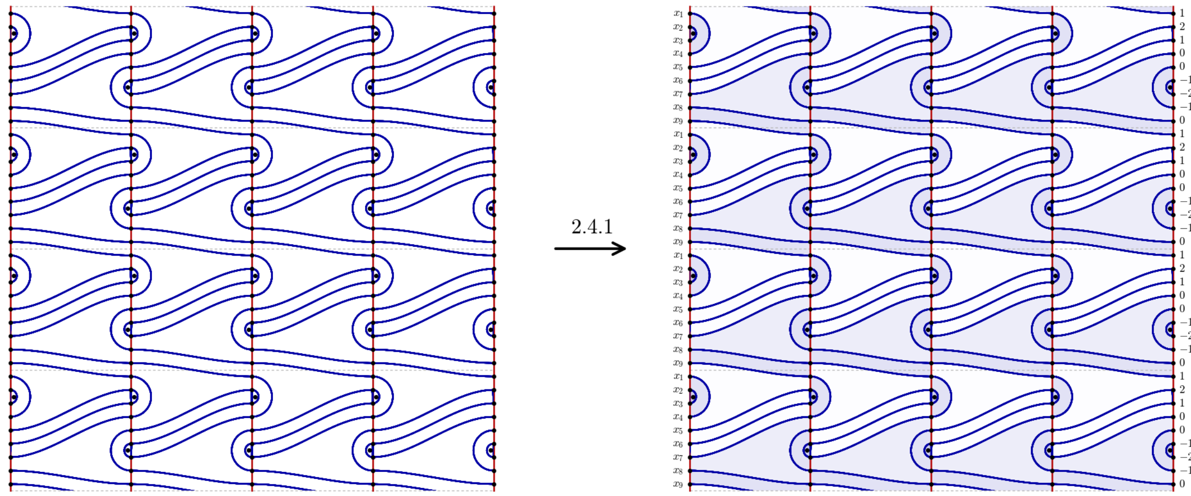
□



(a) $\pi^{-1}(\alpha\cup\beta)\subset\mathbb{R}^2$, $\pi_1(T^2)\cong\mathbb{Z}^2=\langle T_u,T_v\rangle$

(b) $A(x_i)-A(x_j)=n_z(\mathcal{D})-n_w(\mathcal{D})$

FIGURE 12. The cover of $D[7,2,3,1]$, type A, the arrow tinting each complementary face F by $\#(\partial F\cap\alpha)$ as in Definition 2.3.1, so the two per fundamental domain take the darkest wash and $A=(1,0,-1,0,1,0,-1)$.



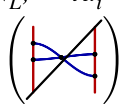
(a) $\pi^{-1}(\alpha\cup\beta)\subset\mathbb{R}^2$, $\pi_1(T^2)\cong\mathbb{Z}^2=\langle T_u,T_v\rangle$ (b) $A(x_i)-A(x_j)=n_z(\mathcal{D})-n_w(\mathcal{D})$

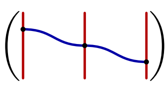
FIGURE 13. The cover of $D[9,2,3,2]$, type B, read the same way through Definition 2.3.1, giving $A=(1,2,1,0,0,-1,-2,-1,0)$.

Lemma 2.6.4 (Lifted strands in the cyclic cover)**.** *Let $\pi\colon \widehat{T^2}\to T^2$ unroll α, with $\widehat{X}$ the lift of a strand X oriented in $+x$ and τ the deck translation by one cell.*

(1) $\widehat{T^2}=\mathbb{R}\times\mathbb{R}/p\mathbb{Z}$, $\quad\widehat{\alpha}=\mathbb{Z}\times\mathbb{R}/p\mathbb{Z}$, $\quad C_k=[k,k+1]\times\mathbb{R}/p\mathbb{Z}$;

(2) $a_i^k=\widehat{R}_{v+i}\in\widehat{\alpha}\cap\partial N_R^k$, $\quad b_j^k=\widehat{L}_{j+1}\in\widehat{\alpha}\cap\partial N_L^k$, $\quad\tau a_i^k=a_i^{k+1}$, $\quad\tau b_j^k=b_j^{k+1}$;

(3) $\widehat{X}\cap\widehat{Y}=\varnothing\ \ (X\neq Y)\iff\beta$ *embedded*: $\Big($  $\Big)=\varnothing$;

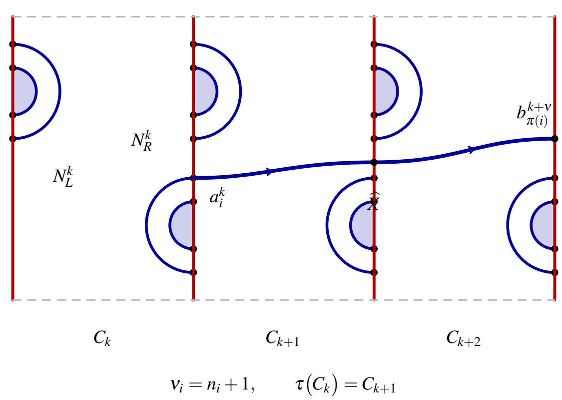
(4) $\nu_i=n_i+1$: $\Big($  $\Big)$.

Proof.

$$\pi^{-1}(\alpha)=\mathbb{Z}\times\mathbb{R}/p\mathbb{Z},\qquad \tau(C_k,N_L^k,N_R^k)=(C_{k+1},N_L^{k+1},N_R^{k+1})\implies(1),(2)$$

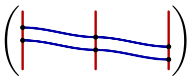
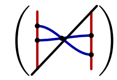
$$\beta\text{ embedded}\iff\widehat{\beta}\text{ embedded}\implies(3)$$

$$N_R^k\xrightarrow{\ \downarrow\downarrow\ }C_{k+1}\xrightarrow{c^{(1)}}\cdots\xrightarrow{c^{(n_i)}}C_{k+n_i+1}\xrightarrow{\ \downarrow\downarrow\ }N_L^{k+n_i+1}\implies\nu_i=n_i+1\implies(4)$$



□

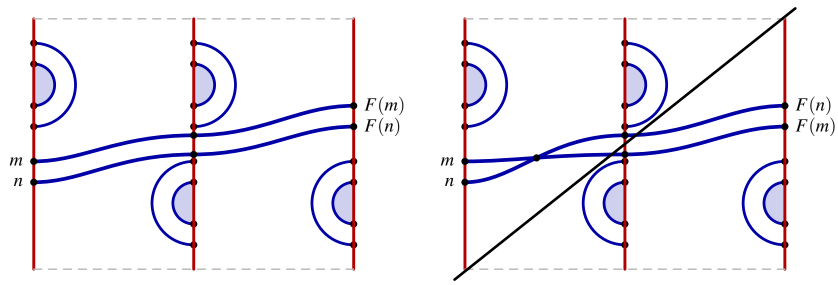
Lemma 2.6.5 (Shift rigidity of the strand matching)**.**

 $\Longrightarrow F\big(4k+i\big)=4\big(k+\nu_i\big)+\pi(i)=4k+i+c \Longleftrightarrow 4\nu_i+\pi(i)-i=c \quad \forall i,$  $=\varnothing.$

Proof.

$$m<n,\ F(m)>F(n) \Longrightarrow \text{interleaved ends} \overset{2.6.4}{\Longrightarrow} \widehat{X}_m \cap \widehat{X}_n \neq \varnothing \Longrightarrow \perp, \qquad F\tau=\tau F \Longrightarrow F(n+4)=F(n)+4$$

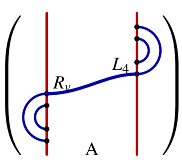
$$F(i+1)-F(i)\geq 1, \qquad \sum_{i=0}^{3}\big(F(i+1)-F(i)\big)=\big(F(0)+4\big)-F(0)=4 \Longrightarrow F(n)=n+c \Longleftrightarrow 4\nu_i+\pi(i)-i=c$$



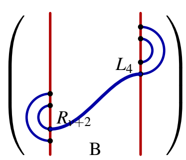
$F(m)<F(n)$ $\qquad$ $F(m)>F(n)$

□

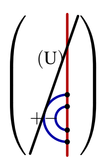
Lemma 2.6.6 (The two mixed diagram types)**.**



or



,



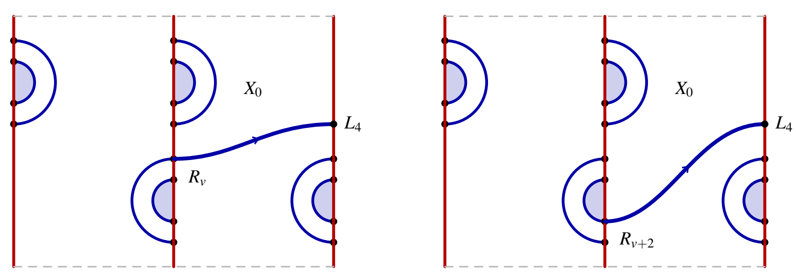
$=\varnothing.$

Proof.

$$\overset{2.6.5}{\Longrightarrow} \pi(i)-i\equiv c \pmod 4\ \forall i, \qquad \text{(U)} \overset{2.6.1}{\Longrightarrow} \big(\pi(i)-i\big)\in\big\{(2,0,-2,0),(0,2,0,-2)\big\} \Longrightarrow \nexists c \Longrightarrow \text{(U)}=\varnothing$$

$$\pi_{\mathrm{A}}=(3,0,1,2) \Longrightarrow \nu_1=\nu_2=\nu_3=\nu_0+1, \qquad \pi_{\mathrm{B}}=(1,2,3,0) \Longrightarrow \nu_0=\nu_1=\nu_2=\nu_3-1, \qquad \overset{2.6.4}{\Longrightarrow} n_i=\nu_i-1$$

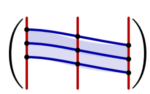
$$\overset{2.5.9}{\Longrightarrow} \big(\eta(O_L),\eta(O_R),\eta(I_L),\eta(I_R)\big)=(-1,-1,+1,+1) \text{ in both}, \qquad \mathrm{A}\colon O_L,O_R,I_L,I_R, \quad \mathrm{B}\colon O_L,I_R,I_L,O_R$$



A $\qquad$ B

□

Lemma 2.6.7 (Product domains between consecutive strands)**.**

$\nu_{j-1}=\nu_j=\nu_{j+1} \Longrightarrow \mathscr{D}_{j-1,j},\mathscr{D}_{j,j+1}=$  *product domains*, $\qquad \alpha\cap\beta\cap\operatorname{int}\mathscr{D}_{j\pm1,j}=\varnothing,$

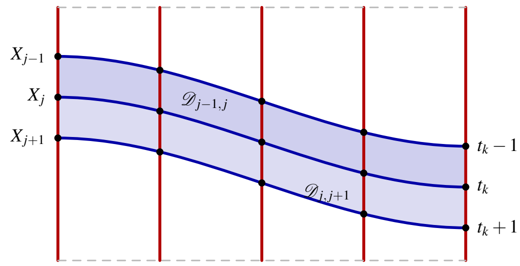
$$\big(X_{j-1},X_j,X_{j+1}\big)=\big(\operatorname{rev}(X_j)-1,\ X_j,\ \operatorname{rev}(X_j)+1\big),\qquad m_{j-1}=m_j=m_{j+1},\qquad j=\begin{cases}2, & \text{A},\\ 1, & \text{B}.\end{cases}$$

Proof.

$$\overset{2.6.6}{\Longrightarrow}\ \nu_{j-1}=\nu_j=\nu_{j+1}\implies \partial\mathscr{D}_{j\pm1,j}\subset\widehat{X}\cup\widehat{\alpha},\qquad x\in\alpha\cap\beta\cap\operatorname{int}\mathscr{D}\implies N_R\hookrightarrow\operatorname{int}\mathscr{D}\implies\perp$$

$$W_L^k\cap\big(X_{j-1}\cup X_j\cup X_{j+1}\big)=\{t_k-1,t_k,t_k+1\}\subset\text{one branch of }\tilde\rho\implies\tilde\rho(t_k+\varepsilon)=\tilde\rho(t_k)+\varepsilon$$

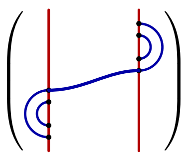
$$\mathbf{1}_{\mathscr{B}}\circ\tilde\rho\text{ constant on each branch}\implies m_{j-1}=m_j=m_{j+1}$$

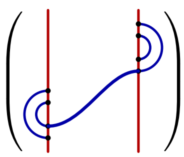


$\tilde\rho(t_k+\varepsilon)=\tilde\rho(t_k)+\varepsilon$

□

Lemma 2.6.8 (Conjugate strand counts)**.**

A =  $\implies \mathbf{n}=N\mathbf{1}-e\,\mathbf{e}_0,\quad \mathbf{m}=m\mathbf{1}-\ell\,\mathbf{e}_0,$

B =  $\implies \mathbf{n}=N\mathbf{1}+e\,\mathbf{e}_3,\quad \mathbf{m}=m\mathbf{1}+\ell\,\mathbf{e}_3.$

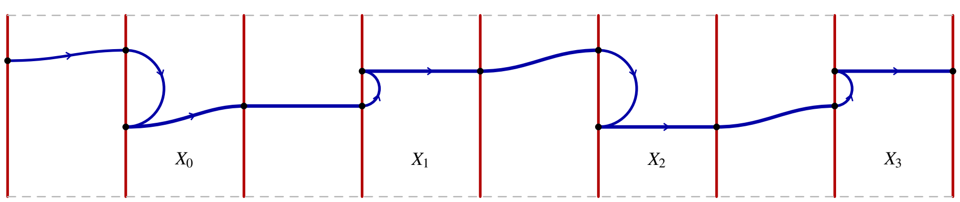
$$\mathbf{n}=N\mathbf{1}-e\,\varepsilon,\qquad \mathbf{m}=m\mathbf{1}-\ell\,\varepsilon,\qquad \varepsilon=\begin{cases}\mathbf{e}_0, & \text{A},\\ -\mathbf{e}_3, & \text{B},\end{cases}\qquad \sigma^\top\varepsilon=-1,\qquad \mathbf{1}^\top\varepsilon=\pm1.$$

Proof.

$$\overset{2.6.1}{\Longrightarrow}\ e=\sum_{s=5}^{p}\chi(c_s)=\sigma^\top\mathbf{n},\quad \ell=\sigma^\top\mathbf{m},\qquad \overset{2.6.6,\,2.6.7}{\Longrightarrow}\ \mathbf{n}-N\mathbf{1},\ \mathbf{m}-m\mathbf{1}\text{ supported on one strand}$$

$$\text{A}:\ e=\sigma^\top\big(N\mathbf{1}+(n_0-N)\mathbf{e}_0\big)=N-n_0\implies\mathbf{n}=N\mathbf{1}-e\,\mathbf{e}_0,\qquad \text{B}:\ e=n_3-N\implies\mathbf{n}=N\mathbf{1}+e\,\mathbf{e}_3$$

$$\text{likewise }\mathbf{m}\text{ with }\ell,\qquad \sigma^\top\mathbf{e}_0=-1,\quad \sigma^\top\mathbf{e}_3=+1\implies\sigma^\top\varepsilon=-1,\quad \mathbf{1}^\top\varepsilon=\pm1$$

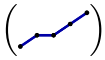
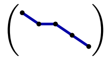


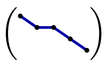
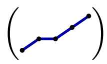
$\mathbf{n}=N\mathbf{1}-e\,\varepsilon,\qquad \mathbf{m}=m\mathbf{1}-\ell\,\varepsilon$

□

Corollary 2.6.9 (Monotonicity of A along a strand)**.** *Lift A to $\widehat{A} = A \circ \pi$ on $\widehat{\alpha} \cap \widehat{\beta}$, so that $d\widehat{A} := \widehat{A}(y) - \widehat{A}(x) = \Theta(c)$ across each arc $\widehat{c}\colon x \to y$ of $\widehat{\beta}$. Then*

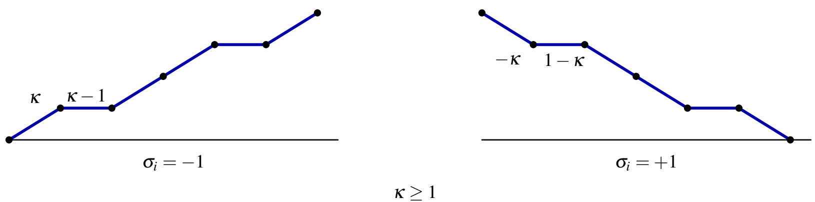
$$d\widehat{A}\big|_{\widehat{X}_i} = \sigma_i\big(\mathbf{1}_{\mathscr{B}} - \kappa\big) \in \sigma_i \cdot \{-\kappa, 1-\kappa\}, \qquad \sigma_i \cdot d\widehat{A}\big|_{\widehat{X}_i} \le 0 \iff \kappa \ge 1,$$

$\kappa \ge 1 \implies \widehat{A}\big|_{X_0,X_2} =$ , $\widehat{A}\big|_{X_1,X_3} =$ ,

$\kappa \le 0 \implies \widehat{A}\big|_{X_0,X_2} =$ , $\widehat{A}\big|_{X_1,X_3} =$ .

Proof.

$$\overset{2.6.3,\,2.6.1}{\Longrightarrow} \sigma_i \cdot \Theta(c_s) = \mathbf{1}_{\mathscr{B}}(c_s) - \kappa \in \{-\kappa, 1-\kappa\}, \qquad \kappa \ge 1 \implies \le 0, \qquad \kappa \le 0 \implies \ge 0$$

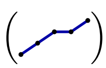
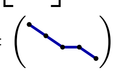


□

Corollary 2.6.10 (Total Alexander increment of a strand)**.**

$$\begin{bmatrix} p-4 \\ b \end{bmatrix} = \begin{bmatrix} \mathbf{1}^\top \mathbf{n} \\ \mathbf{1}^\top \mathbf{m} \end{bmatrix} = \begin{bmatrix} 4N \\ 4m \end{bmatrix} - \big(\mathbf{1}^\top \varepsilon\big) \begin{bmatrix} e \\ \ell \end{bmatrix}, \qquad \mathbf{1}^\top \varepsilon = \begin{cases} +1, & \text{A}, \\ -1, & \text{B}, \end{cases} \qquad S = \kappa N - m$$

$$\mathbf{m} - \kappa \mathbf{n} = -S\mathbf{1} \implies \Theta = \operatorname{diag}(\sigma)\big(\mathbf{m} - \kappa \mathbf{n}\big) = -S\sigma = S \begin{bmatrix} +1 \\ -1 \\ +1 \\ -1 \end{bmatrix}, \qquad \mathbf{1}^\top \Theta = 0,$$

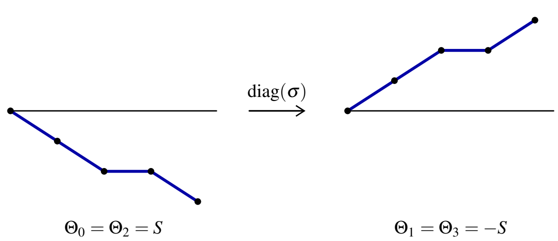
$\sum_{\widehat{X}_0} d\widehat{A} = \sum_{\widehat{X}_2} d\widehat{A} =$  $= S,$ $\qquad \sum_{\widehat{X}_1} d\widehat{A} = \sum_{\widehat{X}_3} d\widehat{A} =$  $= -S.$

Proof.

$$\mathbf{1}^\top \mathbf{n} = \sum_{s=5}^{p} 1 = p - 4, \quad \mathbf{1}^\top \mathbf{m} = b \overset{2.6.8}{\Longrightarrow} \begin{bmatrix} p-4 \\ b \end{bmatrix} = \begin{bmatrix} 4N \\ 4m \end{bmatrix} - \big(\mathbf{1}^\top \varepsilon\big) \begin{bmatrix} e \\ \ell \end{bmatrix}$$

$$\kappa e = e^2 \ell = \ell \implies \mathbf{m} - \kappa \mathbf{n} = (m - \kappa N)\,\mathbf{1} + (\kappa e - \ell)\,\varepsilon = -S\mathbf{1}$$

$$\Theta(X_i) = \sigma_i\big(m_i - \kappa n_i\big) \implies \Theta = \operatorname{diag}(\sigma)\big(\mathbf{m} - \kappa \mathbf{n}\big) = -S\sigma, \qquad \mathbf{1}^\top \Theta = -S\mathbf{1}^\top \sigma = 0$$

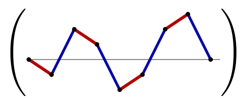
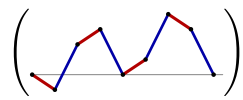


□

Lemma 2.6.11 (The Alexander profile of $\partial\beta$)**.** *Order the eight generators of one period of $\widehat{\beta}$ along* *and let $\widehat{\Theta} \in \mathbb{Z}^8$ be the vector of successive increments of $\widehat{A}$. Then*

$$\widehat{\Theta} = \begin{bmatrix} -1 & S & \mp 1 & -S & +1 & S & \pm 1 & -S \end{bmatrix}^{\top}, \qquad \textstyle\sum_k \widehat{\Theta}_k = 0 \implies \widehat{A} \circ \tau^4 = \widehat{A},$$

and the graph of $\widehat{A}$ over one period, wall steps in the wall colour and strand steps in the rainbow colour, is

A: , B: ,

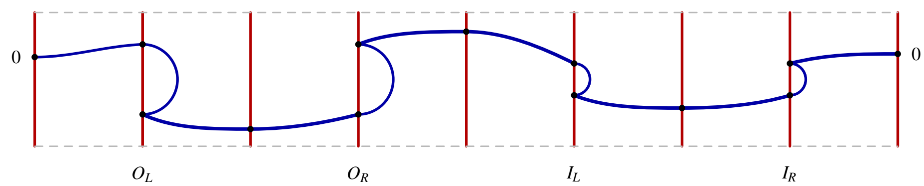
$$\mathrm{A}:\ \ \max \widehat{A} = S \textit{ at one point}, \qquad \mathrm{B}:\ \ \max \widehat{A} = S+1 \textit{ at one point} \qquad (S \geq 1).$$

Proof.

$$\overset{2.5.9,\,2.6.10}{\Longrightarrow} \text{cap steps } \eta(c), \quad \text{strand totals } \pm S, \qquad \overset{2.6.6}{\Longrightarrow} \eta_{\mathrm{A}} = (-1,-1,+1,+1), \quad \eta_{\mathrm{B}} = (-1,+1,+1,-1)$$

$$\widehat{\Theta}_{\mathrm{A}} = \begin{bmatrix} -1 & S & -1 & -S & +1 & S & +1 & -S \end{bmatrix}^{\top}, \qquad \mathrm{A}: 0,-1,S-1,S-2,-2,-1,S-1,S,0$$

$$\widehat{\Theta}_{\mathrm{B}} = \begin{bmatrix} -1 & S & +1 & -S & +1 & S & -1 & -S \end{bmatrix}^{\top}, \qquad \mathrm{B}: 0,-1,S-1,S,0,1,S+1,S,0, \qquad \textstyle\sum_k \widehat{\Theta}_k = 0$$

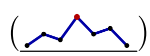


$\widehat{A} \circ \tau^4 = \widehat{A}$

$\square$

Lemma 2.6.12 (Simplicity of the top Alexander grading)**.**

$$S \neq 0 \implies A(\;) = \Big(\;\Big) \implies \mathrm{rk}\widehat{HFK}\big(K, A_{\mathrm{top}}\big) = \#\big\{x \mid A(x) = A_{\mathrm{top}}\big\} = 1.$$



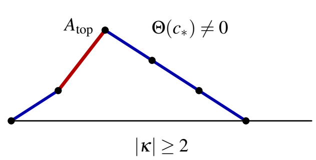
Proof.

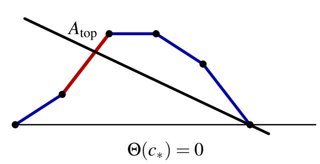
$$\overset{2.6.9}{\Longrightarrow} |\kappa| \geq 2 \implies 0 \notin \{\kappa, \kappa-1\} \implies \widehat{A} \text{ strictly monotone on every strand} \overset{2.6.11}{\Longrightarrow} A_{\mathrm{top}} \text{ attained once}$$

$$\kappa = 1 \implies \ell = 1 \overset{2.2.8}{\Longrightarrow} r = 1 \implies \overline{\mathscr{B}} = \varnothing \implies b = p-4, \quad \mathbf{m} = \mathbf{n}, \quad S = 0 \quad (\text{vacuous})$$

$$\kappa = 0,\ S = -m \neq 0 \overset{2.2.8}{\Longrightarrow} r = p-3, \quad \tilde{\rho}(s) \equiv s-8\ (\mathscr{B}),\ \tilde{\rho}(s) \equiv s-4\ (\overline{\mathscr{B}}), \qquad m \geq 1,\ \ell = 0 \implies b \geq 2$$

$$\overset{2.6.11}{\Longrightarrow} \rho(s_*) \in \{1,2\}, \quad c_* \in \overline{\mathscr{B}} \implies s_* \in \{5,6\} \subset \mathscr{B} \implies \perp \implies c_* \in \mathscr{B}, \quad \big|\Theta(c_*)\big| = 1 \overset{\mathrm{A}}{\Longrightarrow} \mathrm{rk}\widehat{HFK}\big(K, A_{\mathrm{top}}\big) = 1$$

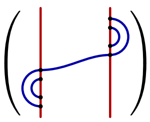




$\square$

FIGURE 14. The splitting $S^3 = V_\alpha \cup_{T^2} V_\beta$, the cut annulus $A = T^2 \smallsetminus \alpha$ carrying the gluing, and the resulting K.

Remark 2.6.13 (What §2.6 computes). Every statement above is phrased in $\mathbf{n}, \mathbf{m}, \sigma$, which are read off one period of $\widehat{\beta}$; isotopy of β in T^2, handle slides over α and stabilisation act on $D[p,2,b,r]$ without changing K, so the chain



$$\xrightarrow{2.6.8} \begin{bmatrix}\mathbf{n} & \mathbf{m}\end{bmatrix} = \begin{bmatrix}N\mathbf{1} & m\mathbf{1}\end{bmatrix} - \varepsilon \cdot \begin{bmatrix}e & \ell\end{bmatrix} \xrightarrow{2.6.10} \Theta = -S\sigma \xrightarrow{2.6.12} \mathrm{rk}\widehat{HFK}(K, A_{\mathrm{top}}) = 1$$

is an invariant of K and not of the tuple that presented it. In particular $S \neq 0$ forces the top group to be one dimensional, so A_{top} is detected by a single generator and

$$S \neq 0 \implies \mathrm{rk}\widehat{HFK}(K, A_{\mathrm{top}}) = 1, \qquad S = 0 \implies \max A - \min A = 2.$$

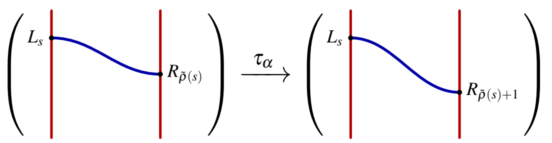
2.7. **Handlebodies, gluing, and the mapping class action.** Behind the splitting of Figure 14 sits one handle decomposition,

$$V = h^0 \cup_{\varphi_1} h^1 \cong S^1 \times D^2, \qquad \varphi_1 : S^0 \times D^2 \hookrightarrow \partial h^0, \qquad D = \{0\} \times D^2, \qquad \partial D = \alpha;$$

compressing disks of a solid torus are isotopic, so all information sits in φ, that is in (α, β, z, w), with $|[\alpha] \cdot [\beta]| = |e|$ as in Definition 2.2.3.

Lemma 2.7.1 (The wall twists r)**.** *Let τ_α be the Dehn twist along α, acting on φ by regluing A after one further turn. Then*

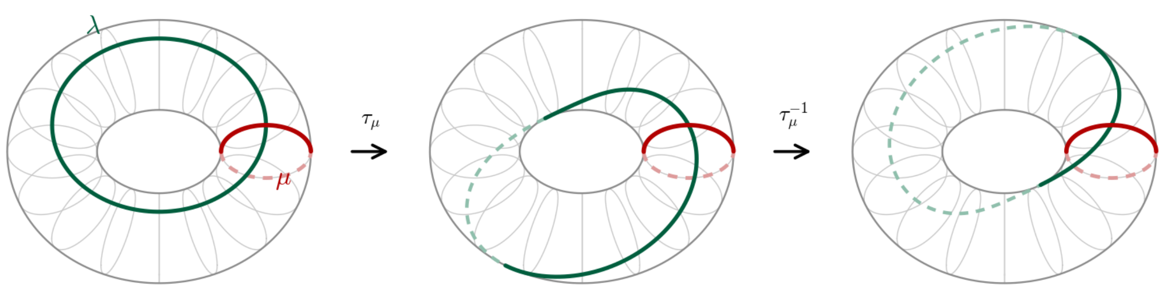
$$\tilde{\rho} \xmapsto{\tau_\alpha} \tilde{\rho} + 1 \iff \tau_\alpha \cdot D[p,2,b,r] = D[p,2,b,r+1], \qquad \langle \tau_\alpha \rangle \cong \mathbb{Z}/p,$$



Proof.

$$\tilde{\rho}(s) - r \text{ and } v - r = b \text{ are independent of } r \implies r \mapsto r+1 \implies \tilde{\rho} \mapsto \tilde{\rho} + 1, \quad \{R_v, \ldots, R_{v+3}\} \mapsto \{R_{v+1}, \ldots, R_{v+4}\}$$
$$\implies \text{the right wall rotates by one foot} = \tau_\alpha \cdot \varphi, \qquad \tilde{\rho} + p \equiv \tilde{\rho} \implies \tau_\alpha^p = \mathrm{id}, \quad \langle \tau_\alpha \rangle \cong \mathbb{Z}/p$$

□



(a) $H_1 = \mathbb{Z}\mu \oplus \mathbb{Z}\lambda$ (b) $\tau_\mu \lambda = \lambda + \mu$ (c) $\tau_\mu^{-1} \lambda = \lambda - \mu$

FIGURE 15. $\mathrm{MCG}(T^2) \cong SL(2,\mathbb{Z})$ acting on $H_1(T^2) = \mathbb{Z}\mu \oplus \mathbb{Z}\lambda$ with $\mu = [\alpha]$, by $\left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$ and its inverse.

Neither condition of Definition 2.2.3 is τ_α-invariant, so a $\mathbb{Z}/p$ orbit leaves the admissible locus, and all three failure modes occur inside one orbit.

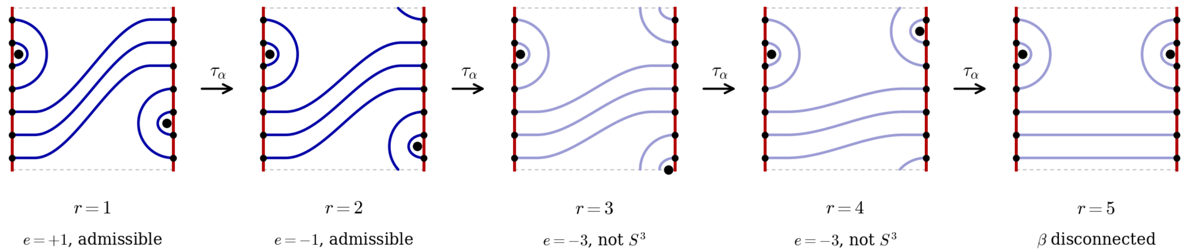


FIGURE 16. The $\langle\tau_\alpha\rangle$-orbit of $D[7,2,3,1]$: the sign of e flips, then $|e|\neq 1$, then β disconnects. Faded panels are not S^3-admissible.

Remark 2.7.2 (Reversal and mirror). The reversal of Convention 2.4.1 is $-I\in SL(2,\mathbb{Z})$, so $\tau_\alpha, -I,$ generate the visible action on tuples, and Corollary 2.4.4 picks $e=+1$ out of each $\pm I$ pair.

What §2.6 counts is holomorphic: the of Corollary 2.6.3 and the product domain of Lemma 2.6.7 are the only embedded index one shapes, each carrying exactly one strip of Definition 4.2.1; see Proposition 4.2.4 and Example 4.2.5.

Example 2.7.3 (Three diagrams and their vectors)**.** From the traversal, via Lemma 2.6.8,

$$[5,2,0,2]:\quad \mathbf{n}=\begin{bmatrix}0\\0\\0\\1\end{bmatrix},\quad \mathbf{m}=\begin{bmatrix}0\\0\\0\\0\end{bmatrix},\quad e=\sigma^\top\mathbf{n}=1,\quad \ell=0,\quad \kappa=0,\quad \Theta=\mathbf{0},\quad S=0,$$

$$[7,2,3,1]:\quad \mathbf{n}=\begin{bmatrix}0\\1\\1\\1\end{bmatrix}=1\cdot\mathbf{1}-1\cdot\mathbf{e}_0,\quad \mathbf{m}=\begin{bmatrix}0\\1\\1\\1\end{bmatrix},\quad \kappa=1,\quad \mathbf{m}-\kappa\mathbf{n}=\mathbf{0},\quad S=0,$$

$$[13,2,6,2]:\quad \mathbf{n}=\begin{bmatrix}2\\2\\2\\3\end{bmatrix}=2\cdot\mathbf{1}+1\cdot\mathbf{e}_3,\quad \mathbf{m}=\begin{bmatrix}2\\2\\2\\0\end{bmatrix}=2\cdot\mathbf{1}-2\cdot\mathbf{e}_3,\quad \kappa=-2,$$

$$\mathbf{m}-\kappa\mathbf{n}=\begin{bmatrix}2\\2\\2\\0\end{bmatrix}+2\cdot\begin{bmatrix}2\\2\\2\\3\end{bmatrix}=\begin{bmatrix}6\\6\\6\\6\end{bmatrix}=-S\mathbf{1},\qquad \Theta=\operatorname{diag}(\sigma)\begin{bmatrix}6\\6\\6\\6\end{bmatrix}=\begin{bmatrix}-6\\+6\\-6\\+6\end{bmatrix},\qquad S=-6.$$

$D[7,2,3,1]$ is the sharp case of Lemma 2.6.12: $S=0$ and A_{top} is attained twice, so $S\neq 0$ cannot be dropped.

3. PROOF OF THEOREM A

Theorem 3.0.1 (Two-rainbow coherence,[37])**.** *Let $[p,2,b,r]$ be S^3-admissible in the sense of Definition 2.2.3 and present $K\subset S^3$, oriented so that O_L runs downwards. Then*

(1) the only embedded bigons of $T^2\setminus(\alpha\cup\beta)$ are $_z$ and $_w$, so

$$\widehat{\partial}=\mathbf{0},\qquad \operatorname{rk}\widehat{HFK}(K,s)=\#\{x\mid A(x)=s\};$$

(2) the caps alternate walls along , cutting β into four strands X_0,X_1,X_2,X_3 with

$$\sigma=\begin{bmatrix}-1 & +1 & -1 & +1\end{bmatrix}^\top,\qquad e=\sigma^\top\mathbf{n},\qquad \ell=\sigma^\top\mathbf{m};$$

(3) *$= \{\,,\,\}$ or $\{\,,\,\}$ $\iff$ D is $\iff$ K is an L-space knot, and then*

$$\operatorname{rk}\widehat{HFK}(K, A_{\text{top}}) = 1;$$

(4) *$= \{\,,\,\}$ $\implies$ K is not an L-space knot, D is of type* A *or* B*, and*

$$\mathbf{n} = N\mathbf{1} - e\,\varepsilon, \quad \mathbf{m} = m\mathbf{1} - \ell\,\varepsilon, \quad \varepsilon = \begin{cases} \mathbf{e}_0, & \text{A}, \\ -\mathbf{e}_3, & \text{B}, \end{cases} \quad \begin{bmatrix} p-4 \\ b \end{bmatrix} = \begin{bmatrix} 4N \\ 4m \end{bmatrix} - (\mathbf{1}^\top \varepsilon) \begin{bmatrix} e \\ \ell \end{bmatrix};$$

(5) *with $\kappa = e\,\ell$ and $S = \kappa N - m$,*

$$\mathbf{m} - \kappa\mathbf{n} = -S\mathbf{1} \implies \Theta = \operatorname{diag}(\sigma)(\mathbf{m} - \kappa\mathbf{n}) = -S\sigma, \qquad \mathbf{1}^\top\Theta = 0 \implies \widehat{A} \circ \tau^4 = \widehat{A};$$

(6) *exactly one of*

$$S \neq 0 \implies \operatorname{rk}\widehat{HFK}(K, A_{\text{top}}) = 1, \qquad S = 0 \implies A(\alpha \cap \beta) = \{-1, 0, +1\}, \quad A_{\text{top}} = 1, \quad g(K) = 1;$$

(7) *the second case is exactly the two ends of the band count,*

$$S = 0 \iff \mathbf{1}_{\mathscr{B}} \equiv \kappa \in \{0, 1\} \iff b \in \{0,\ p-4\};$$

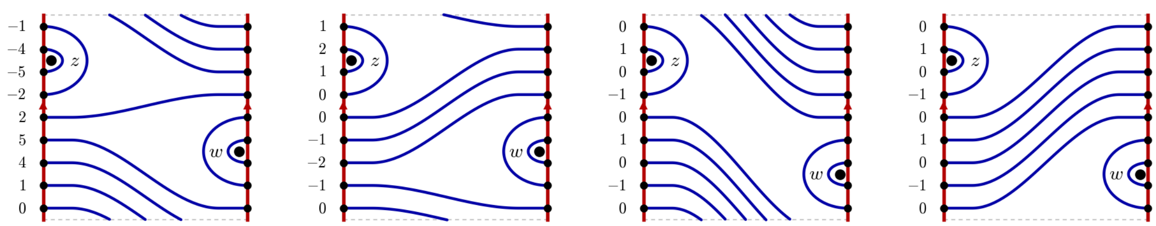
(8) *the Dehn twist along α acts by*

$$\tau_\alpha \cdot D[p, 2, b, r] = D[p, 2, b, r+1], \qquad \langle \tau_\alpha \rangle \cong \mathbb{Z}/p,$$

and (1)–(7) *are invariants of K, not of the tuple.*

In particular

$$\operatorname{rk}\widehat{HFK}(K, A_{\text{top}}) = 1 \qquad \text{or} \qquad A_{\text{top}} = 1.$$



$D[9,2,1,4]$, $(-,-,-,-)$ — (3) coherent, $A_{\text{top}} = 5$

$D[9,2,3,2]$, $(-,+,+,-)$ — (6) $S \neq 0$, $A_{\text{top}} = 2$

$D[9,2,0,6]$, $(-,+,+,-)$ — (7) $S = 0$, $b = 0$, $A_{\text{top}} = 1$

$D[9,2,5,1]$, $(-,+,+,-)$ — (7) $S = 0$, $b = p-4$, $A_{\text{top}} = 1$

FIGURE 17. The four cases of Theorem 3.0.1 at $p = 9$, each drawn on the annulus $T^2 \setminus \alpha$ whose two red edges are glued by , with the sign word $(\eta(O_L), \eta(O_R), \eta(I_L), \eta(I_R))$ printed beneath and the number beside a foot equal to $A(x_h)$.

Proof. (1),

$$\text{admissible} \overset{2.3.6}{\Longrightarrow} \ \in \{\ _z, \ _w\}, \qquad \widehat{\partial} = \mathbf{0} \implies \operatorname{rk}\widehat{HFK}(K, s) = \#\{x \mid A(x) = s\}$$

(2),

$$\text{nested} \overset{2.6.1}{\Longrightarrow} \text{alternate} \implies \sigma = \begin{bmatrix} -1 & +1 & -1 & +1 \end{bmatrix}^\top$$

(3),

$$= \{\,,\,\}, \{\,,\,\} \overset{2.3.8}{\iff} \ \overset{2.5.1}{\iff} K \text{ an L-space knot} \overset{2.5.2}{\Longrightarrow} \operatorname{rk}\widehat{HFK}(K, A_{\text{top}}) = 1$$

(4),

$$= \{\,,\,\} \overset{2.3.8}{\Longrightarrow} \neg \ \overset{2.5.1}{\iff} K \text{ not an L-space knot}, \qquad \overset{2.6.6}{\Longrightarrow} \text{A or B}$$

$$\overset{2.6.8}{\Longrightarrow} \ \mathbf{n} = N\mathbf{1} - e\,\varepsilon, \quad \mathbf{m} = m\mathbf{1} - \ell\,\varepsilon \overset{2.6.10}{\Longrightarrow} \begin{bmatrix} p-4 \\ b \end{bmatrix} = \begin{bmatrix} \mathbf{1}^\top \mathbf{n} \\ \mathbf{1}^\top \mathbf{m} \end{bmatrix}$$

(5),

$$\kappa e = e^2 \ell = \ell \overset{2.6.10}{\Longrightarrow} \mathbf{m} - \kappa \mathbf{n} = -S\mathbf{1}, \quad \Theta = -S\sigma, \quad \mathbf{1}^\top \Theta = -S\mathbf{1}^\top \sigma = 0 \overset{2.6.11}{\Longrightarrow} \widehat{A} \circ \tau^4 = \widehat{A}$$

(6),

$$S \neq 0 \overset{2.6.12}{\Longrightarrow} \mathrm{rk}\widehat{HFK}\big(K, A_{\mathrm{top}}\big) = 1, \qquad S = 0 \overset{2.6.10,\,2.6.9}{\Longrightarrow} d\widehat{A}\big|_{X_i} = 0 \overset{2.6.11}{\Longrightarrow} A(\alpha \cap \beta) = \{-1, 0, +1\}$$

$$\Longrightarrow A_{\mathrm{top}} = 1, \quad \widehat{HFK}\big(K, A_{\mathrm{top}}\big) \neq 0 \Longrightarrow g(K) = \max\big\{s \mid \widehat{HFK}(K,s) \neq 0\big\} = 1$$

by genus detection [24]: a one signed sum vanishes only termwise, and the three strand levels are centred by $A = -A \circ \mathrm{conj}$. (7),

$$d\widehat{A}\big|_{X_i} = \sigma_i\big(\mathbf{1}_{\mathscr{B}} - \kappa\big) \Longrightarrow \Big[S = 0 \iff \mathbf{1}_{\mathscr{B}} \equiv \kappa \in \{0,1\} \iff \mathbf{m} \in \{\mathbf{0}, \mathbf{n}\} \iff b \in \{0,\ p-4\}\Big]$$

(8),

$$\overset{2.7.1}{\Longrightarrow} \tau_\alpha \cdot D[p,2,b,r] = D[p,2,b,r+1], \qquad \tau_\alpha^p = \mathrm{id} \overset{2.6.13}{\Longrightarrow} \text{invariants of } K$$

The last line is (3) and (6) together. □

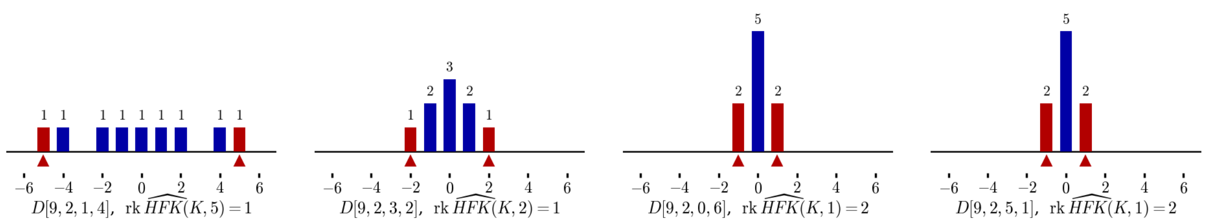


FIGURE 18. The groups $\mathrm{rk}\widehat{HFK}(K,s)$ of the four diagrams of Figure 17 on one pair of axes, the columns and the marks at $s = \pm A_{\mathrm{top}}$ in red, so that the gaps of the staircase, the single top of $S \neq 0$ and the three levels of $S = 0$ are read side by side.

Remark 3.0.2 (The two ends of the band count). By 3.0.1,

$$\Delta_K(t) = \pm\Big[(N+1)\big(t + t^{-1}\big) - \big(2N + 2 + \varepsilon\big)\Big], \qquad \varepsilon = \begin{cases} +1, & \text{B}, \\ -1, & \text{A}, \end{cases} \qquad N = \left\lfloor \tfrac{p-3}{4} \right\rfloor.$$

4. Analytic foundations and classification input

4.1. Detection theorems.

Theorem 4.1.1 (Euler characteristic, [23, Equation (1)]).

$$\sum_{s \in \mathbb{Z}} \chi\big(\widehat{HFK}(K,s)\big) \cdot t^s = \Delta_K(t).$$

Proof. [23, Equation (1)]; see also [27]. □

Theorem 4.1.2 (Genus detection and symmetry, [24, Theorem 1.2]).

$$g(K) = \max\big\{s \in \mathbb{Z} \mid \widehat{HFK}(K,s) \neq 0\big\}, \qquad \widehat{HFK}_M(K,s) \cong \widehat{HFK}_{M-2s}(K,-s).$$

Proof. [24, Theorem 1.2] and [23, §3]. □

Theorem 4.1.3 (Fiberedness detection, [31, Theorem 1.1], [33]).

$$K \textit{ fibered} \iff \mathrm{rk}\widehat{HFK}\big(K, g(K)\big) = 1 \Longrightarrow \Delta_K \textit{ monic}.$$

Proof. [31, Theorem 1.1], the genus one case in [33]; monicity by Theorem 4.1.1 at $s = g(K)$. □

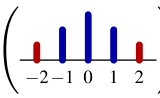
Theorem 4.1.4 (Alternating knots, [21, Theorem 1.3], [22, Theorem 1.4])**.**

$$K \textit{ alternating}, \quad \Delta_K(t) = \sum_s a_s \cdot t^s \implies \mathrm{rk}\widehat{HFK}_M(K,s) = \begin{cases} |a_s|, & M = s + \frac{\sigma(K)}{2}, \\ 0, & \textit{otherwise}, \end{cases} \qquad \tau(K) = -\frac{\sigma(K)}{2}.$$

Proof. [21, Theorem 1.3], [22, Theorem 1.4]. □

Example 4.1.5 (Detection at $D[9,2,3,2]$)**.** From the second panel of Figure 17, via Corollary 2.3.7, Theorem 4.1.1, Theorem 4.1.2 and Theorem 4.1.3,

$$\mathrm{rk}\widehat{HFK}(K,*) = \left(\vphantom{\Big|} \right) = (1,2,3,2,1), \qquad \sum_s \mathrm{rk} = 9 = p,$$



$$\sum_{s\in\mathbb{Z}} \chi\big(\widehat{HFK}(K,s)\big) \cdot t^s = -t^2 + 2t - 1 + 2t^{-1} - t^{-2} = \Delta_K(t), \qquad \Delta_K(1) = 1,$$

$$g(K) = 2 = A_{\mathrm{top}}, \qquad \mathrm{rk}\widehat{HFK}(K,2) = 1 \implies K \text{ fibered}, \quad \Delta_K \text{ monic}.$$

At $s = 0$ three generators cancel to $\chi = -1$: Theorem 4.1.1 is strictly weaker than Corollary 2.3.7.

Example 4.1.6 (The thin case $D[5,2,1,1]$)**.** For the diagram of Figure 11, by Theorem 4.1.4 with $\sigma(K) = 0$,

$$\Delta_K(t) = -t + 3 - t^{-1}, \qquad \big(\mathrm{rk}\widehat{HFK}(K,s)\big)_{s=1,0,-1} = (1,3,1) = \big(|a_1|, |a_0|, |a_{-1}|\big),$$

$$\widehat{HFK}_M(K,s) \neq 0 \implies M = s + \tfrac{\sigma(K)}{2} = s, \qquad \tau(K) = -\tfrac{\sigma(K)}{2} = 0:$$

this is 4_1, the first entry of Remark 3.0.2, revisited in Example 4.3.3.

4.2. **The nonlinear Cauchy–Riemann problem on the strip.** Every count in §2.6 is a count of solutions of a nonlinear elliptic boundary value problem with totally real boundary conditions on the strip $\mathbb{R} \times [0,1]$. The problem is stated in Definition 4.2.1, linearised in Theorem 4.2.2, compactified in Theorem 4.2.3, and solved for the two domains of Corollary 2.6.3 and Lemma 2.6.7 in Proposition 4.2.4.

Definition 4.2.1 (Floer strip)**.** For $x, y \in \alpha \cap \beta$, $\phi \in \pi_2(x,y)$ and J an almost complex structure on $\mathrm{Sym}^1(T^2) = T^2$,

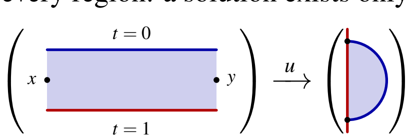
$$\mathscr{M}_J(\phi) = \Big\{ u\colon \mathbb{R} \times [0,1] \to T^2 \;\Big|\; \partial_s u + J(u)\,\partial_t u = 0,\ u(\mathbb{R},0) \subset \beta,\ u(\mathbb{R},1) \subset \alpha,\ [u] = \phi \Big\},$$

$$\lim_{s\to-\infty} u(s,\cdot) = x, \qquad \lim_{s\to+\infty} u(s,\cdot) = y, \qquad |u(s,t) - x| = O\big(e^{-\delta|s|}\big), \quad \delta > 0.$$

The equation is the $\overline{\partial}$-operator of the complex structure J, and α, β are totally real in $\mathrm{Sym}^1(T^2) = T^2$, so u is smooth up to the boundary and the energy is the area of the domain,

$$E(u) = \tfrac{1}{2} \int_{\mathbb{R}\times[0,1]} |du|^2\, ds\, dt = \int_{\mathbb{R}\times[0,1]} u^*\omega = \mathrm{Area}\big(\mathscr{D}(\phi)\big) = \sum_{R\in\pi_0(T^2\setminus(\alpha\cup\beta))} n_R(\phi) \cdot \mathrm{Area}(R),$$

so $E(u) \geq 0$ forces $n_R(\phi) \geq 0$ for every region: a solution exists only for a nonnegative domain,



Theorem 4.2.2 (Fredholm index and transversality, [23, §3], [29, §4])**.** *For $p>2$ and $\delta>0$ smaller than the smallest angle at x,y,*

$$D_u\overline{\partial}_J\colon W^{1,p}_\delta\big(u^*TT^2;\,\alpha,\beta\big)\longrightarrow L^p_\delta\big(\Omega^{0,1}\otimes u^*TT^2\big),\qquad D_u\overline{\partial}_J\,\xi=\nabla_s\xi+J(u)\nabla_t\xi+\big(\nabla_\xi J\big)\,\partial_t u$$

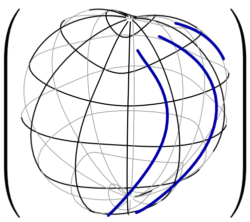
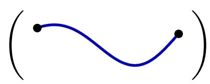
is Fredholm, and Riemann–Roch computes its index as the Maslov index, a combinatorial quantity in the domain [29, Corollary 4.10]*,*

$$\operatorname{ind}D_u\overline{\partial}_J=\mu(\phi)=e\big(\mathscr{D}(\phi)\big)+n_x\big(\mathscr{D}(\phi)\big)+n_y\big(\mathscr{D}(\phi)\big),\qquad e(\mathscr{D})=\chi(\mathscr{D})-\tfrac14\big(\#corners\big),$$

n_x,n_y being the averages of the four local multiplicities at x and at y. The universal moduli space is cut out transversally, so by Sard–Smale a generic J makes every $\mathscr{M}_J(\phi)$ a smooth manifold with

$$\dim\mathscr{M}_J(\phi)=\mu(\phi),\qquad \mathbb{R}\curvearrowright\mathscr{M}_J(\phi)\ \textit{freely by}\ (\sigma\cdot u)(s,t)=u(s-\sigma,t),\qquad \widehat{\mathscr{M}}(\phi)=\mathscr{M}_J(\phi)/\mathbb{R};$$

at $\mu(\phi)=2$,

 $\xrightarrow{/\mathbb{R}}$ , $\qquad \dim\mathscr{M}_J(\phi)=2,\qquad \dim\widehat{\mathscr{M}}(\phi)=1,$

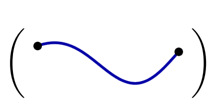
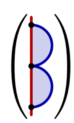
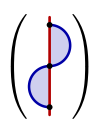
the flow lines being the orbits of the translation.

Proof. [23, §3], [29, §4]; the linear theory as in [25, Appendix C]. □

Theorem 4.2.3 (Gromov compactness, [23, §3], [29, §7])**.**

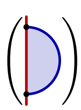
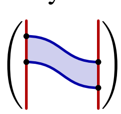
$$E(u)=\operatorname{Area}\big(\mathscr{D}(\phi)\big)<\infty\ \textit{depends only on}\ \phi\implies\textit{a priori bound on}\ \|du\|_{L^2},$$
$$\pi_2(T^2)=0,\quad n_z(\phi)=0\implies\textit{no sphere and no boundary bubbling},$$
$$\mu(\phi)=1\implies\widehat{\mathscr{M}}(\phi)\ \textit{a compact}\ 0\textit{-manifold},$$
$$\mu(\phi)=2\implies\partial\widehat{\mathscr{M}}(\phi)=\bigsqcup_{\substack{\phi_1*\phi_2=\phi\\ \mu(\phi_1)=\mu(\phi_2)=1}}\widehat{\mathscr{M}}(\phi_1)\times\widehat{\mathscr{M}}(\phi_2)\implies\widehat{\partial}^2=\mathbf{0},$$

the ends of the quotient being the broken strips,

∂  $=$  $\sqcup$  .

Proof. [23, §3], [29, §7]; compactness as in [25, Chapter 4]. □

Proposition 4.2.4 (Automatic regularity of the two shapes)**.**

$\mathscr{D}(\phi)=$  *or*  *embedded* $\implies\#\widehat{\mathscr{M}}(\phi)=1$ *for every J.*

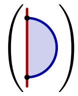
Proof.

$\mathscr{D}(\phi)$ embedded $\implies\mathscr{M}_J(\phi)=\big\{$biholomorphisms $\mathbb{R}\times[0,1]\to\mathscr{D}(\phi)\big\}$ by the Riemann mapping theorem

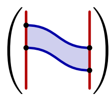
$$\dim_{\mathbb{C}}T^2=1\implies\operatorname{coker}D_u\overline{\partial}_J=0\ \ \forall J\implies\#\widehat{\mathscr{M}}(\phi)=\#\big(\operatorname{Aut}(\mathbb{R}\times[0,1])/\mathbb{R}\big)=1\quad[28,\S2],[30,\S3]$$

The two shapes are exactly Corollary 2.6.3 and Lemma 2.6.7, so every count in §2.6 is a count of solutions of Definition 4.2.1. □

Example 4.2.5 (The index of the two shapes)**.** A disk with $2n$ convex corners has $e = \chi - \frac{2n}{4} = 1 - \frac{n}{2}$, and a generator at which $\mathscr{D}(\phi)$ occupies k of its four quadrants contributes $\frac{k}{4}$; Theorem 4.2.2 then reads



$$: \quad \mu = \underbrace{1 - \tfrac{2}{4}}_{e} + \underbrace{\tfrac{1}{4}}_{n_x} + \underbrace{\tfrac{1}{4}}_{n_y} = \tfrac{1}{2} + \tfrac{1}{2} = 1,$$



$$: \quad \mu = \underbrace{1 - \tfrac{4}{4}}_{e} + \underbrace{\tfrac{2}{4}}_{n_x} + \underbrace{\tfrac{2}{4}}_{n_y} = 0 + 1 = 1.$$

On the diagram of Figure 11 the bigon $_z$ has $n_z = 1$, so it is excluded from $\widehat{\partial}$ by Definition 4.2.1; every other embedded bigon is excluded by Theorem 2.3.6, and $\widehat{\partial} = \mathbf{0}$ as in Corollary 2.3.7.

4.3. **The genus-one classification.**

Theorem 4.3.1 (Schubert classification, [12], [13, Chapter 12])**.** *For p, p' odd,* $\gcd(p,q) = \gcd(p',q') = 1$, $0 < q < p$, $0 < q' < p'$,

$$\mathfrak{b}(p,q) \cong \mathfrak{b}(p',q') \iff p = p', \quad q' \equiv q^{\pm 1} \pmod{p}, \qquad \overline{\mathfrak{b}(p,q)} \cong \mathfrak{b}(p, p-q),$$

and every $\mathfrak{b}(p,q)$ is a $(1,1)$-knot.

Proof. [12], [13, Chapter 12]; the $(1,1)$ property in [28, §1]. □

Theorem 4.3.2 (Genus one two-bridge knots, [13, Prop. 12.25], [20, §1])**.** *K is genus one and two-bridge $\iff$ $K \cong J(2m, 2\varepsilon n)$ with $m,n > 0$, $\varepsilon = \pm 1$, and then*

$$J(2m, 2\varepsilon n) = \mathfrak{b}\big(4mn - \varepsilon,\ 4mn - 2n - \varepsilon\big), \qquad V = \begin{bmatrix} m & 1 \\ 0 & \varepsilon n \end{bmatrix},$$

$$\det\big(V - t \cdot V^{\top}\big) = \det \begin{bmatrix} m(1-t) & 1 \\ -t & \varepsilon n(1-t) \end{bmatrix} = \varepsilon mn(1-t)^2 + t = \varepsilon mn t^2 + (1 - 2\varepsilon mn)t + \varepsilon mn,$$

$$\Delta_K(t) = t^{-1} \cdot \det\big(V - t \cdot V^{\top}\big) = \varepsilon mn\big(t + t^{-1}\big) + 1 - 2\varepsilon mn, \qquad \sigma(K) = 1 + \varepsilon,$$

$$\big(\mathrm{rk}\widehat{HFK}(K,1),\ \mathrm{rk}\widehat{HFK}(K,0),\ \mathrm{rk}\widehat{HFK}(K,-1)\big) = \big(mn,\ 2mn - \varepsilon,\ mn\big).$$

Proof. [13, Prop. 12.25], [20, §1], [19]; two-bridge knots are alternating [13, Chapter 12], so the ranks are $|a_s|$ by Theorem 4.1.4. □

Example 4.3.3 (Two tabulated rows as double twist knots)**.** At $(m,n,\varepsilon) = (1,1,-1)$, Theorem 4.3.2 gives

$$J(2,-2) = \mathfrak{b}(4+1,\ 4-2+1) = \mathfrak{b}(5,3), \qquad V = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}, \qquad V - t \cdot V^{\top} = \begin{bmatrix} 1-t & 1 \\ -t & t-1 \end{bmatrix},$$

$$\det\big(V - t \cdot V^{\top}\big) = (1-t)(t-1) - 1 \cdot (-t) = -t^2 + 3t - 1 \implies \Delta_K(t) = -t + 3 - t^{-1}, \qquad \sigma(K) = 0,$$

$$\big(\mathrm{rk}\widehat{HFK}(K,1), \mathrm{rk}\widehat{HFK}(K,0), \mathrm{rk}\widehat{HFK}(K,-1)\big) = \big(mn,\ 2mn - \varepsilon,\ mn\big) = (1,3,1),$$

which is $D[5,2,1,1]$ of Figure 11 and Example 4.1.6. At $(m,n,\varepsilon) = (1,2,-1)$,

$$J(2,-4) = \mathfrak{b}(8+1,\ 8-4+1) = \mathfrak{b}(9,5), \qquad \Delta_K(t) = -2t + 5 - 2t^{-1}, \qquad \big(\mathrm{rk}\widehat{HFK}\big) = (2,5,2),$$

which is both $S = 0$ rows at $p = 9$ – the last two panels of Figure 17 and of Figure 18. These are 4_1 and 6_1 of Remark 3.0.2; Theorem 4.3.7 and Theorem 4.3.4 show they exhaust the possibilities there.

Theorem 4.3.4 (Hatcher–Thurston, [14])**.**

$$K \textit{ two-bridge} \implies S^3 \setminus \operatorname{int} \nu(K) \textit{ atoroidal} \implies K \textit{ not a nontrivial satellite}.$$

Proof. [14]. □

Theorem 4.3.5 (Morimoto–Sakuma, [15])**.** *The satellite $(1,1)$-knots are exactly the Morimoto–Sakuma family; each has tunnel number one and a nontrivial torus knot companion.*

Proof. [15]. □

Theorem 4.3.6 (Satellite genus, [11])**.** *For $K = P(C)$ with pattern $P \subset V$, winding number w and companion C,*

$$g(K) = |w| \cdot g(C) + g_V(P), \qquad g(K) = 1, \quad g(C) \geq 1, \quad P \not\simeq \operatorname{core}(V) \implies w = 0.$$

Proof.

$$1 = |w| \cdot g(C) + g_V(P), \quad g(C) \geq 1 \implies \big(|w|, g_V(P)\big) \in \{(1,0),(0,1)\}$$
$$|w| = 1,\ g_V(P) = 0 \implies \exists \text{ annulus } A \subset V,\ \partial A = P \sqcup \lambda \implies P \simeq \operatorname{core}(V) \implies \perp \implies w = 0$$

□

Theorem 4.3.7 (Matsuda, [16])**.**

$$K \textit{ genus one } (1,1) \iff K \cong J(2m,2n),\ mn \neq 0, \quad \textit{or} \quad K \in \mathscr{MS}_1,$$

where $\mathscr{MS}_1$ is the genus one Morimoto–Sakuma family.

Proof. [16], [15], with Theorem 4.3.2 and Theorem 4.3.5; see also [36, §1] for the $(1,1)$ satellites and [40, 41] for the two-bridge side. □

Corollary 4.3.8 (Where $S = 0$ lands)**.**

$$S = 0 \overset{3.0.1}{\Longrightarrow} A_{\text{top}} = 1 \overset{4.1.2}{\Longrightarrow} g(K) = 1 \overset{4.3.7}{\Longrightarrow} K \cong J(2m,2n) \ \textit{ or } \ K \in \mathscr{MS}_1,$$

and in the second case Theorem 4.3.6 forces $w(P) = 0$.

Proof. Theorem 3.0.1 (6), Theorem 4.1.2, Theorem 4.3.7, Theorem 4.3.6. □

Theorem 4.3.9 (Whitehead doubles, [32, Theorems 1.1–1.2])**.** *$\widehat{HFK}\big(D_\pm(C,t)\big)$ is determined by t and the filtered chain homotopy type of $\widehat{CFK}(C)$, and*

$$\tau\big(D_+(C,t)\big) = \begin{cases} 1, & t < 2\tau(C), \\ 0, & t \geq 2\tau(C); \end{cases} \qquad \big\{D_\pm(U,t)\big\}_{t \in \mathbb{Z}} = \{\textit{twist knots}\} \cup \{U\}.$$

Proof. [32, Theorems 1.1–1.2]. □

4.4. **Parametrization and regular homotopy.**

Theorem 4.4.1 (Winding numbers on surfaces, [1], [3])**.** *For S compact oriented, $v \in \Gamma(TS)$ nowhere zero, and regular immersions $\gamma_0, \gamma_1 : S^1 \looparrowright S$,*

$$\gamma_0 \simeq_{\text{reg}} \gamma_1 \iff [\gamma_0]_{\text{free}} = [\gamma_1]_{\text{free}} \ \textit{ and } \ \operatorname{wind}_v(\gamma_0) = \operatorname{wind}_v(\gamma_1),$$
$$S = \mathbb{R}^2 \implies \big[\gamma_0 \simeq_{\text{reg}} \gamma_1 \iff \operatorname{rot}(\gamma_0) = \operatorname{rot}(\gamma_1)\big].$$

Proof. [1] for the plane, [3] in general; see [5, §1.2]. □

This is the analytic fact behind Lemma 2.3.5: an embedded essential $\beta \subset T^2$ is regularly homotopic to a geodesic, so $\vartheta(\beta) = 0$ and the cap senses balance, which is what forces $\sigma = \begin{bmatrix} -1 & +1 & -1 & +1 \end{bmatrix}^\top$ in Lemma 2.6.1 and hence $e = \sigma^\top \mathbf{n}$ in Lemma 2.6.8. See also [2], [4].

Theorem 4.4.2 (Mapping class parametrization, [18])**.**

$$\operatorname{Mod}\big(T^2, \{z,w\}\big) \twoheadrightarrow \big\{(1,1)\textit{-knots}\big\}/\cong, \qquad \varphi \longmapsto K_\varphi, \qquad D_\varphi = \big(T^2, \alpha, \varphi(\beta), z, w\big).$$

Proof. [18]. □

Under this surjection the orbit of Lemma 2.7.1 is the image of $\langle \tau_\alpha \rangle \cong \mathbb{Z}/p$ in $\mathrm{MCG}(T^2) \cong SL(2,\mathbb{Z})$ of Figure 15, and Figure 16 is the corresponding walk through the fibres; the admissible locus is not τ_α-invariant, so the walk leaves it after finitely many steps. Compare [17, 38, 39] for other parametrizations of the same set.

Lemma 4.4.3 (Finite disk bound, [37, Prop. 3.1])**.** *For every tuple $D[p,a,b,r]$ the set of embedded Whitney disks is finite, and*

$$\#\big(\mathscr{D} \cap \widetilde{\beta}\big) \le \left\lfloor \tfrac{p}{2} \right\rfloor.$$

Proof. [35], [37, Prop. 3.1]. □

4.5. **The Alexander cocycle and the dynamics of the traversal.**

Definition 4.5.1 (Return map, cocycle)**.** On $\alpha \cap \beta = \{x_1, \dots, x_p\} \cong \mathbb{Z}/p$, ordered by wall height,

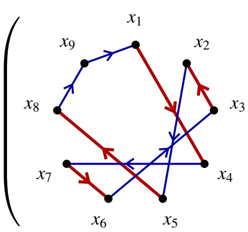
$$F = \omega \circ \sigma \colon \mathbb{Z}/p \longrightarrow \mathbb{Z}/p, \qquad \Theta(h) := \Theta(c_h) = \begin{cases} \eta(c_h), & c_h \text{ a cap}, \\ \chi(c_h)\big(\mathbf{1}_{\mathscr{B}}(c_h) - \kappa\big), & c_h \text{ a crossing arc}, \end{cases}$$

where c_h is the outgoing arc of at x_h (Definition 2.2.1, Corollary 2.6.3, Lemma 2.5.9).

Lemma 4.5.2 (Eight branches)**.**

$$F(h) - h \in \{\pm 1,\ \pm 3\} \cup \{\pm(r-5),\ \pm(r-1)\} \pmod{p}, \qquad \Longleftrightarrow \ F \text{ a } p\text{-cycle},$$

$$F\big(D[9,2,3,2]\big) = \left(\vphantom{\rule{0pt}{60pt}} \right.$$



$$\left. \vphantom{\rule{0pt}{60pt}} \right) = (147632589),$$

cap steps in the wall colour, crossing steps in the rainbow colorr.

Proof.

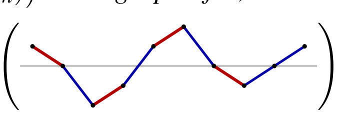
$$\overset{2.2.5}{\Longrightarrow} F(h) - h = \Delta h(c_h) \in \{\pm 1, \pm 3\} \cup \{\pm(r-5), \pm(r-1)\}, \qquad F \text{ a } p\text{-cycle} \iff |\,\,| = p \overset{2.2.7}{\Longleftrightarrow}$$

□

Theorem 4.5.3 (Transfer function, [6])**.** *For admissible $D[p,2,b,r]$,*

$$\sum_{h \in \mathbb{Z}/p} \Theta(h) = 0, \qquad A \circ F - A = \Theta, \qquad \varphi \circ F - \varphi = \Theta \implies \varphi = A + \text{const},$$

the constant fixed by $A = -A \circ \mathrm{conj}$; the orbit of the skew product $\widetilde{F}(h,k) = \big(F(h), k + \Theta(h)\big)$ through $\big(h, A(x_h)\big)$ is the graph of A,



$$= \mathrm{graph} A \textit{ of } D[9,2,3,2], \quad \textit{one period, closing to } A(x_1) = 1.$$

Proof.

$$\overset{2.6.10}{\Longrightarrow} \sum_h \Theta(h) = \mathbf{1}^\top \Theta + \sum_{c \text{ cap}} \eta(c) = 0 + 0 = 0, \qquad \overset{2.5.8}{\Longrightarrow} A\big(F(h)\big) - A(h) = \Theta(c_h) = \Theta(h)$$

$$(\varphi - A) \circ F = \varphi - A \overset{4.5.2}{\Longrightarrow} \varphi - A \text{ constant on the single } F\text{-orbit}$$

For minimal systems, solvability $\iff$ bounded Birkhoff sums is [6]; on a finite orbit the obstruction is exactly the mean. □

Theorem 4.5.4 (Oscillation equals drift)**.** *For mixed $D[p,2,b,r]$, normalised as in Corollary 2.4.4,*

$$\max A - \min A = |S| + 2, \qquad A_{\mathrm{top}} = \tfrac{1}{2}|S| + 1, \qquad S \in 2\mathbb{Z},$$

$$g(K) = \tfrac{1}{2}\,|\kappa N - m| + 1, \qquad \mathrm{rk}\widehat{HFK}\big(K, g(K)\big) = \begin{cases} 1, & S \neq 0, \\ N+1, & S = 0. \end{cases}$$

Proof.

$$\overset{2.6.9}{\Longrightarrow} \text{ extrema of } A \text{ at the eight cap nodes } \overset{2.6.11}{\Longrightarrow} \max A - \min A = |S| + 2 \quad \forall S \in \mathbb{Z}, \text{ both types}$$

$$A = -A \circ \mathrm{conj} \implies A_{\mathrm{top}} = \tfrac{1}{2}\big(\max A - \min A\big) = \tfrac{1}{2}|S| + 1 \in \mathbb{Z} \implies S \in 2\mathbb{Z} \overset{2.3.6,\,4.1.2}{\Longrightarrow} g(K) = A_{\mathrm{top}}$$

$$S \neq 0 \overset{2.6.12}{\Longrightarrow} \mathrm{rk}\widehat{HFK}\big(K, A_{\mathrm{top}}\big) = 1, \qquad S = 0 \overset{2.6.10,\,2.6.9}{\Longrightarrow} A \text{ constant on strands}$$

$$\implies \big\{A = A_{\mathrm{top}}\big\} = \text{one strand and its two cap feet } \overset{2.6.8}{\Longrightarrow} \mathrm{rk}\widehat{HFK}\big(K, A_{\mathrm{top}}\big) = N + 1$$

□

Corollary 4.5.5 (Fiberedness)**.**

$$K \textit{ fibered} \iff \mathrm{rk}\widehat{HFK}\big(K, g(K)\big) = 1 \iff S \neq 0 \ \textit{ or } \ N = 0,$$

and then Δ_K is monic; the unique fibered member with $S = 0$ is 4_1 at $p = 4N + 5 = 5$.

Proof. Theorem 4.1.3, Theorem 4.5.4; $N = 0$ forces type B and $p = 5$ by Corollary 2.6.10; monicity is Theorem 4.1.1 at $s = g(K)$. □

4.6. **The tabulation.**

Definition 4.6.1 (Tabulation)**.**

$$\mathscr{T}_p = \big\{\, D[p,2,b,r] \;\big|\; \,\big\}, \qquad \mathscr{T} = \bigcup_p \mathscr{T}_p,$$

the S^3-admissible tuples of Definition 2.2.3.

Remark 4.6.2 (Growth). $|\mathscr{T}_p| = 8, 6, 12, 12, 18, 16, 22$ for $p = 5, 7, \ldots, 17$, ninety four in all, split forty eight against forty six mixed; each p contributes the mixed row $D[p,2,0,p-3]$, so $\mathscr{T}$ presents infinitely many knot types.

Proof.

$$D[p,2,0,p-3] \in \mathscr{T}_p \overset{3.0.2}{\Longrightarrow} \Delta_K = \pm\Big[(N+1)\big(t + t^{-1}\big) - \big(2N + 2 + \varepsilon\big)\Big], \qquad N = \Big\lfloor \tfrac{p-3}{4} \Big\rfloor \text{ unbounded}$$

□

p,b,r	type	ℓ	κ	N	m	S	A_{top}	$\#A_{\mathrm{top}}$	span
5, 0, 2	B	0	0	0	0	0	1	1	2
5, 1, 1	B	1	1	0	0	0	1	1	2
7, 0, 4	A	0	0	1	0	0	1	2	2
7, 3, 1	A	1	1	1	1	0	1	2	2
9, 0, 6	B	0	0	1	0	0	1	2	2
9, 3, 2	B	−1	−1	1	1	−2	2	1	4
9, 5, 1	B	1	1	1	1	0	1	2	2
11, 0, 8	A	0	0	2	0	0	1	3	2
11, 1, 4	A	−1	−1	2	0	−2	2	1	4
11, 7, 1	A	1	1	2	2	0	1	3	2
13, 0, 10	B	0	0	2	0	0	1	3	2
13, 6, 2	B	−2	−2	2	2	−6	4	1	8
13, 8, 10	B	0	0	2	2	−2	2	1	4
13, 9, 1	B	1	1	2	2	0	1	3	2
15, 0, 12	A	0	0	3	0	0	1	4	2
15, 2, 4	A	−2	−2	3	0	−6	4	1	8
15, 8, 12	A	0	0	3	2	−2	2	1	4
15, 11, 1	A	1	1	3	3	0	1	4	2
17, 0, 14	B	0	0	3	0	0	1	4	2
17, 3, 10	B	−1	−1	3	1	−4	3	1	6
17, 8, 14	B	0	0	3	2	−2	2	1	4
17, 9, 2	B	−3	−3	3	3	−12	7	1	14
17, 13, 1	B	1	1	3	3	0	1	4	2

FIGURE 19. The normalised tabulation: the mixed tuples with $e=+1$ and $p \le 17$, with type, ℓ, κ, N, m, $S=\kappa N-m$, A_{top}, its multiplicity, and span$A$; the identities $A_{\mathrm{top}}=\frac{1}{2}|S|+1$ and $\operatorname{span}A=|S|+2$ of Theorem 4.5.4 hold in every row.

APPENDIX A. FOUNDATIONAL TOOLS

A.1. **Permutation matrices and cycles.** A permutation π of $\{1,\dots,d\}$ acts by $P_\pi \cdot \mathbf{e}_n = \mathbf{e}_{\pi(n)}$, one 1 per row and per column, and

$$P_\pi \cdot P_\tau = P_{\pi\tau}, \qquad P_\pi^{-1} = P_\pi^{\top}, \qquad \det\bigl(P_\pi - \lambda \mathbf{I}_d\bigr) = \prod_{i=1}^{k}(-1)^{d_i}\bigl(\lambda^{d_i}-1\bigr), \qquad \dim\ker\bigl(P_\pi - \mathbf{I}_d\bigr) = k,$$

for π a disjoint union of k cycles of lengths $d_1,\dots,d_k$: a nonzero Leibniz term takes $-\lambda$ or 1 from every column of a cycle, and a vector fixed by P_π is constant on each cycle.

Example A.1.1 (The 4-cycle)**.**

$$\pi = (1\,2\,3\,4), \qquad P_\pi = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \qquad \det\bigl(P_\pi - \lambda \mathbf{I}_4\bigr) = \lambda^4 - 1, \qquad \ker\bigl(P_\pi - \mathbf{I}_4\bigr) = \langle \mathbf{1} \rangle.$$

Lemma 2.2.6 applies this with $d = 2p$ and $k = 2$, $d_1 = d_2 = p$, and Lemma 4.5.2 reads the same cycle as the return map F.

A.2. **The cohomological equation on a cycle.** For F a d-cycle on a finite set X and $\theta\colon X \to \mathbb{Z}$, the *cohomological equation* is $\varphi \circ F - \varphi = \theta$, and

$$\exists\,\varphi \iff \sum_{x\in X}\theta(x) = 0, \qquad \varphi\bigl(F^k x_0\bigr) = \varphi(x_0) + \sum_{j=0}^{k-1}\theta\bigl(F^j x_0\bigr), \qquad \varphi \text{ unique up to a constant},$$

existence by telescoping, uniqueness because an F-invariant function is constant on the single orbit. The *oscillation* $\max\varphi - \min\varphi$ is the deviation of the partial sums, and does not depend on the choice of constant.

Example A.2.1 (On $\mathbb{Z}/4$)**.**

$$\theta = (2,-1,-2,1), \qquad \sum \theta = 0, \qquad \varphi = (0,2,1,-1)+c, \qquad \max\varphi - \min\varphi = 3.$$

§4.5 applies this with F the traversal, $\theta = \Theta$ and $\varphi = A$; [6] is the compact minimal analogue, where existence is equivalent to bounded Birkhoff sums.

Appendix B. Reproducibility

B.1. **The engine.**

Listing 1. `tworainbow.py`: the normal form, the traversal, drift, admissibility, geometry and gradings of $D[p,2,b,r]$; every number quoted in the paper is computed here at render time.

```
import math

A = 2


def mod(h, p):
    return (h - 1) % p + 1


def matching(p, b, r, a=A):
    arcs = []
    for i in range(1, a + 1):
        arcs.append((("L", mod(i, p)), ("L", mod(2 * a + 1 - i, p))))
        arcs.append((("R", mod(r + b - 1 + i, p)),
                     ("R", mod(r + b + 2 * a - i, p))))
    for s in range(2 * a + 1, p + 1):
        rho = r + s - 2 * a - 1 if s <= 2 * a + b else r + s - 1
        arcs.append((("L", mod(s, p)), ("R", mod(rho, p))))
    return arcs


def traverse(p, b, r, a=A):
    arcs = matching(p, b, r, a)
    ends = [e for arc in arcs for e in arc]
    if len(set(ends)) != 2 * p:
        return None
    nxt = {}
    for u, v in arcs:
        nxt[u], nxt[v] = v, u
    walk, cur = [], ("L", 1)
    for _ in range(p):
        head = nxt[cur]
        walk.append((frozenset((cur, head)), cur, head))
        side, h = head
        cur = ("R" if side == "L" else "L", h)
    if cur != ("L", 1) or len({w[0] for w in walk}) != p:
        return None
    return walk


def invariants(p, b, r, a=A):
    walk = traverse(p, b, r, a)
    if walk is None:
        return None
    v = r + b
    true = {}
    for i in range(1, a + 1):
```

```
        true[("L", mod(i, p))] = i
        true[("L", mod(2 * a + 1 - i, p))] = 2 * a + 1 - i
        true[("R", mod(v - 1 + i, p))] = v - 1 + i
        true[("R", mod(v + 2 * a - i, p))] = v + 2 * a - i

    drift = 0
    caps = []
    blocks, cur = [], []
    for k, (_, tail, head) in enumerate(walk):
        if tail[0] == head[0]:

            sense = 1 if true[head] < true[tail] else -1
            caps.append((k, tail[0], sense))
            blocks.append(cur)
            cur = []
        else:
            chi = 1 if tail[0] == "L" else -1
            drift += chi
            cur.append((k, chi, tail, head))
    if cur:
        if blocks:
            blocks[0] = cur + blocks[0]
        else:
            blocks = [cur]
    if abs(drift) != 1:
        return None

    uL = sum(1 for _, wall, s in caps if wall == "L" and s == 1)
    uR = sum(1 for _, wall, s in caps if wall == "R" and s == 1)
    dL, dR = a - uL, a - uR
    assert uL - dL == uR - dR, (p, b, r, uL, uR)
    return {"p": p, "b": b, "r": r, "drift": drift, "caps": caps,
            "uL": uL, "uR": uR, "blocks": blocks,
            "word": "".join("+" if s == 1 else "-" for _, _, s in caps),
            "sides": "".join(w for _, w, _ in caps)}


def tabulate(pmax=17, a=A):
    out = []
    for p in range(3, pmax + 1, 2):
        for b in range(0, p - 2 * a + 1):
            for r in range(1, p + 1):
                inv = invariants(p, b, r, a)
                if inv is not None:
                    out.append(inv)
    return out


if __name__ == "__main__":
    C = tabulate()
    print(f"admissible [p,2,b,r] with p <= 17 : {len(C)}")
    by_p = {}
    for t in C:
        by_p.setdefault(t["p"], []).append(t)
    print("  by p:", {p: len(v) for p, v in sorted(by_p.items())})

    bad = [t for t in C if t["uL"] != t["uR"]]
    print(f"turning balance u_L = u_R : {len(C) - len(bad)}/{len(C)} "
          f"(failures {len(bad)})")

    vals = sorted({(t["uL"], t["uR"]) for t in C})
    print("  (u_L,u_R) observed:", vals)
```

```
    sides = sorted({t["sides"] for t in C})
    print("  cap side words:", sides)

    coh = [t for t in C if t["uL"] in (0, 2)]
    mix = [t for t in C if t["uL"] == 1]
    print(f"coherent {len(coh)}   mixed {len(mix)}")
    print("  coherent cap words:", sorted({t['word'] for t in coh}))
    print("  mixed cap words:   ", sorted({t['word'] for t in mix}))

    nblk = sorted({len(t["blocks"]) for t in C})
    print("  blocks per diagram:", nblk)


D_OUT, D_IN = 0.215, 0.095
U0, U1 = 0.14, 0.86


def _smooth(x):
    x = min(1.0, max(0.0, x))
    return x * x * (3 - 2 * x)


def geometry(p, b, r, a=A):
    v = r + b
    deps = [D_OUT - (D_OUT - D_IN) * (i - 1) / max(a - 1, 1)
            for i in range(1, a + 1)]
    out = {}

    def rain(wall, h1, h2, dep, n=60):
        ym, amp = 0.5 * (h1 + h2) - 0.5, 0.5 * (h2 - h1)
        pts = [(dep * math.sin(math.pi * i / n) if wall == "L"
                else 1.0 - dep * math.sin(math.pi * i / n),
                ym - amp * math.cos(math.pi * i / n)) for i in range(n + 1)]
        key = frozenset(((wall, mod(h1, p)), (wall, mod(h2, p))))
        out[key] = ((wall, mod(h1, p)), pts)

    for i in range(1, a + 1):
        rain("L", i, 2 * a + 1 - i, deps[i - 1])
        rain("R", v - 1 + i, v + 2 * a - i, deps[i - 1])
    for s in range(2 * a + 1, p + 1):
        rho = r + s - 2 * a - 1 if s <= 2 * a + b else r + s - 1
        y0, y1 = s - 0.5, rho - 0.5
        pts = [(i / 120, y0 + (y1 - y0) * _smooth((i / 120 - U0) / (U1 - U0)))
               for i in range(121)]
        key = frozenset((("L", mod(s, p)), ("R", mod(rho, p))))
        out[key] = (("L", mod(s, p)), pts)
    zq = (0.42 * D_IN, float(a))
    wq = (1.0 - 0.42 * D_IN, (v + a - 1 - 0.5) % p + 0.5)
    return out, zq, wq


def _cross(a1, a2, b1, b2):
    def cr(o, x, y):
        return (x[0] - o[0]) * (y[1] - o[1]) - (x[1] - o[1]) * (y[0] - o[0])
    d1, d2 = cr(b1, b2, a1), cr(b1, b2, a2)
    d3, d4 = cr(a1, a2, b1), cr(a1, a2, b2)
    if (d1 > 0) != (d2 > 0) and (d3 > 0) != (d4 > 0):
        return 1 if cr(a1, a2, b2) > 0 else -1
    return 0


def _pair(curve, delta, p):
    tot = 0
```

```
    for k in (-2, -1, 0, 1, 2):
        for i in range(len(curve) - 1):
            c1 = (curve[i][0], curve[i][1] + k * p)
            c2 = (curve[i + 1][0], curve[i + 1][1] + k * p)
            for j in range(len(delta) - 1):
                tot += _cross(c1, c2, delta[j], delta[j + 1])
    return tot


_EPS = None


def gradings(p, b, r, a=A):
    walk = traverse(p, b, r, a)
    if walk is None:
        return None
    geo, zq, wq = geometry(p, b, r, a)
    drift = sum(1 if t[0] == "L" else -1 for _, t, h in walk if t[0] != h[0])
    delta = [(zq[0] + 0.0031, zq[1] + 0.0173), (0.5, 0.5 * (zq[1] + wq[1]) + 0.037),
             (wq[0] - 0.0031, wq[1] + 0.0119)]

    steps = []
    for key, tail, head in walk:
        start, pts = geo[key]
        if start != tail:
            pts = pts[::-1]
        chi = 0 if tail[0] == head[0] else (1 if tail[0] == "L" else -1)
        steps.append((tail, head, chi, _pair(pts, delta, p)))
    beta_d = sum(s[3] for s in steps)

    A_ = {}
    cur = 0.0
    for tail, head, chi, ixn in steps:
        if tail[1] not in A_:
            A_[tail[1]] = cur
        cur += _EPS * (ixn - drift * chi * beta_d)
    assert abs(cur) < 1e-9, (p, b, r, cur)
    vals = A_
    shift = 0.5 * (max(vals.values()) + min(vals.values()))
    vals = {h: v - shift for h, v in vals.items()}
    ints = {h: round(v) for h, v in vals.items()}
    assert all(abs(vals[h] - ints[h]) < 1e-6 for h in ints), (p, b, r)
    assert sorted(ints.values()) == sorted(-x for x in ints.values()), \
        (p, b, r, sorted(ints.values()))
    sgn = {}
    for _, head, _, _ in steps:
        sgn[head[1]] = 1 if head[0] == "R" else -1
    dpoly = {}
    for h in ints:
        dpoly[ints[h]] = dpoly.get(ints[h], 0) + sgn[h]
    if sum(dpoly.values()) < 0:
        dpoly = {k: -v for k, v in dpoly.items()}
    assert abs(sum(dpoly.values())) == 1, (p, b, r, dpoly)
    return {"A": ints, "drift": drift, "delta": dpoly}


def _calibrate():
    global _EPS
    for eps in (1, -1):
        _EPS = eps
        ok = True
        for (p, b, r) in [(3, 1, 1), (5, 2, 2)]:
            g = gradings(p, b, r, a=1)
```

```
            assert g is not None, (p, b, r)
            top = max(g["A"], key=lambda h: g["A"][h])
            v = r + b
            zc = {1, 2}
            wc = {mod(v, p), mod(v + 1, p)}
            want = zc if g["drift"] > 0 else wc
            if top not in want:
                ok = False
                break
        if ok:
            return
    raise AssertionError("calibration failed")


_calibrate()
```

B.2. The tabulation checks.

LISTING 2. `chk26.py`: the four-strand matching over the tabulation – cap wall word, χ per block, π by type (Lemma 2.6.6), the shift constant of Lemma 2.6.5, the paired-return identities of Lemma 2.6.8, block totals and the gate of Lemma 2.6.12.

```
from collections import Counter
import tworainbow as T

PI_A = (3, 0, 1, 2)
PI_B = (1, 2, 3, 0)


def analyse(p, b, r):
    walk = T.traverse(p, b, r)
    if walk is None:
        return None
    v = r + b
    true = {}
    for i in (1, 2):
        true[("L", T.mod(i, p))] = i
        true[("L", T.mod(5 - i, p))] = 5 - i
        true[("R", T.mod(v - 1 + i, p))] = v - 1 + i
        true[("R", T.mod(v + 4 - i, p))] = v + 4 - i

    def name(t, h):
        hs = {t[1], h[1]}
        if t[0] == "L":
            return "OL" if hs == {T.mod(1, p), T.mod(4, p)} else "IL"
        return "OR" if hs == {T.mod(v, p), T.mod(v + 3, p)} else "IR"

    k0 = [k for k, (_, t, h) in enumerate(walk)
          if t[0] == h[0] and name(t, h) == "OL"][0]
    w = walk[k0:] + walk[:k0]

    caps, blocks, cur = [], [], []
    for _, t, h in w:
        if t[0] == h[0]:
            caps.append((name(t, h), t, h, 1 if true[h] < true[t] else -1))
            blocks.append(cur)
            cur = []
        else:
            s = None
            L = t if t[0] == "L" else h
            for ss in range(5, p + 1):
```

```
                if T.mod(ss, p) == L[1]:
                    rho = r + ss - 5 if ss <= 4 + b else r + ss - 1
                    R = h if t[0] == "L" else t
                    if T.mod(rho, p) == R[1]:
                        s = ss
                        break
            cur.append((1 if t[0] == "L" else -1, s is not None and s <= 4 + b))
    blocks = blocks[1:] + [cur]
    if len(blocks) != 4 or len(caps) != 4:
        return None
    e = sum(c for B in blocks for c, _ in B)
    if abs(e) != 1:
        return None
    ell = sum(c for B in blocks for c, ib in B if ib)

    order = "A" if caps[1][0] == "OR" else "B"
    du = "D" if caps[1 if order == "A" else 3][3] == -1 else "U"
    walls = "".join("L" if n in ("OL", "IL") else "R" for n, _, _, _ in caps)
    word = "".join("+" if s == 1 else "-" for _, _, _, s in caps)


    def aidx(pt):
        return (true[pt] - v) % p

    def bidx(pt):
        return true[pt] - 1

    pi, nu = {}, {}
    names = ["P1", "Q1", "P2", "Q2"]
    for j, B in enumerate(blocks):

        c0, c1 = caps[j], caps[(j + 1) % 4]
        if names[j][0] == "P":
            bpt, apt = c0[2], c1[1]
        else:
            apt, bpt = c0[2], c1[1]
        i = aidx(apt)
        pi[i] = bidx(bpt)
        nu[i] = len(B) + 1
    if sorted(pi) != [0, 1, 2, 3] or sorted(pi.values()) != [0, 1, 2, 3]:
        return dict(bad="pi not a bijection", p=p, b=b, r=r, pi=pi)
    cs = {4 * nu[i] + pi[i] - i for i in range(4)}

    n = {names[j]: len(B) for j, B in enumerate(blocks)}
    m = {names[j]: sum(1 for _, ib in B if ib) for j, B in enumerate(blocks)}
    return dict(p=p, b=b, r=r, e=e, ell=ell, order=order, du=du,
                walls=walls, word=word, pi=tuple(pi[i] for i in range(4)),
                nu=tuple(nu[i] for i in range(4)), cs=cs, n=n, m=m,
                chis=tuple(tuple(sorted({c for c, _ in B})) for B in blocks))


rows = [analyse(p, b, r) for p in range(5, 18, 2)
        for b in range(0, p - 3) for r in range(1, p + 1)]
rows = [x for x in rows if x and "bad" not in x]
mix = [x for x in rows if x["word"].count("+") == 2 and x["word"] != "++--"
       and x["word"][0] == "-" and set(x["word"]) == {"+", "-"}
       and x["word"] in ("--++", "-++-")]
coh = [x for x in rows if x["word"] in ("----", "++++")]

print(f"total {len(rows)}   mixed {len(mix)}   coherent {len(coh)}")
print("(1) cap wall word:", Counter(x["walls"] for x in mix))
print("(2) chi per block:", Counter(x["chis"] for x in mix))
print("(3) order/DU split:", Counter((x["order"], x["du"]) for x in mix))
```

```
print("    pi by order:  ",
      Counter((x["order"], x["pi"]) for x in mix))
badpi = [x for x in mix
         if x["pi"] != (PI_A if x["order"] == "A" else PI_B)]
print("    pi failures:", len(badpi))
print("(4) 4 nu_i + pi(i) - i constant:",
      sum(1 for x in mix if len(x["cs"]) == 1), "/", len(mix))
print("    c mod 4 by order:",
      Counter((x["order"], next(iter(x["cs"])) % 4) for x in mix))

bad = []
for x in mix:
    n, m, e, l, o = x["n"], x["m"], x["e"], x["ell"], x["order"]
    if o == "A":
        N, M = n["Q2"], m["Q2"]
        ok = (n["Q2"] == n["P2"] == n["Q1"] and m["Q2"] == m["P2"] == m["Q1"]
              and n["P1"] - n["P2"] == -e and m["P1"] - m["P2"] == -l
              and x["p"] == 4 * N + 4 - e and x["b"] == 4 * M - l)
    else:
        N, M = n["P2"], m["P2"]
        ok = (n["P2"] == n["Q1"] == n["P1"] and m["P2"] == m["Q1"] == m["P1"]
              and n["Q1"] - n["Q2"] == -e and m["Q1"] - m["Q2"] == -l
              and x["p"] == 4 * N + 4 + e and x["b"] == 4 * M + l)
    if not ok:
        bad.append(x)
print("(5) paired / block-total identities, failures:", len(bad))

blk = []
for x in mix:
    k = x["e"] * x["ell"]
    N = x["n"]["Q2"] if x["order"] == "A" else x["n"]["P2"]
    M = x["m"]["Q2"] if x["order"] == "A" else x["m"]["P2"]
    S = k * N - M
    Th = {X: (1 if X.startswith("Q") else -1) * (x["m"][X] - k * x["n"][X])
          for X in ("P1", "Q1", "P2", "Q2")}
    if not (Th["P1"] == Th["P2"] == S and Th["Q1"] == Th["Q2"] == -S):
        blk.append((x, S, Th))
print("    block totals Theta(P_i)=S, Theta(Q_i)=-S, failures:", len(blk))

gate = []
for x in mix:
    k = x["e"] * x["ell"]
    N = x["n"]["Q2"] if x["order"] == "A" else x["n"]["P2"]
    M = x["m"]["Q2"] if x["order"] == "A" else x["m"]["P2"]
    S = k * N - M
    g = T.gradings(x["p"], x["b"], x["r"])["A"]
    top = max(g.values())
    mult = sum(1 for u in g.values() if u == top)
    span = max(g.values()) - min(g.values())
    if S != 0 and mult != 1:
        gate.append(("S!=0 but multiple tops", x["p"], x["b"], x["r"], S, mult))
    if S == 0 and span != 2:
        gate.append(("S=0 but span", x["p"], x["b"], x["r"], span))
print("(6) gate lemma + span, failures:", len(gate), gate[:3])
print("    kappa values:", Counter(x["e"] * x["ell"] for x in mix))
print("    S values:", dict(sorted(Counter(
    (x["e"] * x["ell"]) * (x["n"]["Q2"] if x["order"] == "A" else x["n"]["P2"])
    - (x["m"]["Q2"] if x["order"] == "A" else x["m"]["P2"])
    for x in mix).items())))
```

B.3. The vector identities.

LISTING 3. `chkvec2.py`: $\mathbf{n} = N\mathbf{1} - e\,\varepsilon$, $\mathbf{m} = m\mathbf{1} - \ell\,\varepsilon$, $\mathbf{m} - \kappa\mathbf{n} = -S\mathbf{1}$ and $\Theta = -S\sigma$ (Lemma 2.6.8, Corollary 2.6.10) on all forty six mixed tuples.

```python
import chk26 as C

SIG = (-1, 1, -1, 1)
NAMES = ("P1", "Q1", "P2", "Q2")
dot = lambda u, v: sum(a * b for a, b in zip(u, v))

fail = {k: 0 for k in ("functional", "eps", "shift", "theta", "onetheta")}
seen = 0
for x in C.mix:
    seen += 1
    n = tuple(x["n"][k] for k in NAMES)
    m = tuple(x["m"][k] for k in NAMES)
    e, ell, o = x["e"], x["ell"], x["order"]
    kap = e * ell

    if not (dot(SIG, n) == e and dot(SIG, m) == ell
            and sum(n) + 4 == x["p"] and sum(m) == x["b"]):
        fail["functional"] += 1

    eps = (1, 0, 0, 0) if o == "A" else (0, 0, 0, -1)
    N = n[1] if o == "A" else n[0]
    M = m[1] if o == "A" else m[0]
    if not (dot(SIG, eps) == -1 and abs(sum(eps)) == 1):
        fail["eps"] += 1
    if any(n[i] != N - e * eps[i] for i in range(4)) or \
       any(m[i] != M - ell * eps[i] for i in range(4)):
        fail["shift"] += 1

    S = kap * N - M
    if any(m[i] - kap * n[i] != -S for i in range(4)):
        fail["theta"] += 1
    Th = tuple(SIG[i] * (m[i] - kap * n[i]) for i in range(4))
    if Th != tuple(-S * s for s in SIG) or sum(Th) != 0:
        fail["onetheta"] += 1

print(f"mixed tuples checked: {seen}")
for k, v in fail.items():
    print(f"  {k:<12} failures: {v}")
```

B.4. **The statement checks.**

LISTING 4. `chk27.py`: checks (1)–(9) – Figure 19, the vectors of Example 2.7.3, span and parity, the extreme rows of Remark 3.0.2, $\tau_\alpha \colon r \mapsto r+1$ (Lemma 2.7.1), the orbit of Figure 16, the dichotomy (Theorem 3.0.1 (6)–(7)), the four drawn cases of Figure 17, and §4.5.

```python
import re
import chk26 as C
import tworainbow as T

NAMES = ("P1", "Q1", "P2", "Q2")


def scalars(x):
    o, e, ell = x["order"], x["e"], x["ell"]
    kappa = e * ell
    N = x["n"]["Q2"] if o == "A" else x["n"]["P2"]
    m = x["m"]["Q2"] if o == "A" else x["m"]["P2"]
    g = T.gradings(x["p"], x["b"], x["r"])
```

```
    A = g["A"]
    top = max(A.values())
    return dict(order=o, e=e, ell=ell, kappa=kappa, N=N, m=m,
                S=kappa * N - m, top=top,
                cnt=sum(1 for v in A.values() if v == top),
                span=top - min(A.values()), delta=g["delta"])


def find(p, b, r):
    return next(z for z in C.mix if (z["p"], z["b"], z["r"]) == (p, b, r))


import tab2 as V
rows = V.rows()
bad = 0
want = sorted((x["p"], x["b"], x["r"]) for x in C.mix if x["e"] == 1)
if sorted((q["p"], q["b"], q["r"]) for q in rows) != want:
    bad += 1
    print("  row set mismatch")
for q in rows:
    d = scalars(find(q["p"], q["b"], q["r"]))
    got = (d["order"], d["ell"], d["kappa"], d["N"], d["m"], d["S"],
           d["top"], d["cnt"], d["span"])
    drawn = (q["order"], q["ell"], q["kappa"], q["N"], q["m"], q["S"],
             q["top"], q["cnt"], q["span"])
    if d["e"] != 1 or got != drawn:
        bad += 1
        print("  row mismatch", (q["p"], q["b"], q["r"]), drawn, got)
print(f"(1) fig_tab2: {len(rows)} rows, mismatches {bad}")


bad = 0
for tgt, vn, vm in (((5, 0, 2), [0, 0, 0, 1], [0, 0, 0, 0]),
                    ((7, 3, 1), [0, 1, 1, 1], [0, 1, 1, 1]),
                    ((13, 6, 2), [2, 2, 2, 3], [2, 2, 2, 0])):
    x = find(*tgt)
    if [x["n"][k] for k in NAMES] != vn or [x["m"][k] for k in NAMES] != vm:
        bad += 1
        print("  vector mismatch", tgt)
print(f"(2) Example 2.7.1 vectors: mismatches {bad}")


f1 = f2 = f3 = f4 = 0
for x in C.mix:
    d = scalars(x)
    if d["S"] % 2:
        f1 += 1
    if d["top"] != abs(d["S"]) // 2 + 1:
        f2 += 1
    if d["span"] != 2 * d["top"]:
        f3 += 1
    if d["S"] == 0 and d["cnt"] != d["N"] + 1:
        f4 += 1
print(f"(3) tabulation identities over {len(C.mix)} mixed tuples: "
      f"S even {f1}, A_top {f2}, span {f3}, rank at S=0 {f4}")


bad, seen = 0, 0
for x in C.mix:
    p, b = x["p"], x["b"]
    if x["e"] != 1 or b not in (0, p - 4):
        continue
    seen += 1
```

```
    d = scalars(x)
    N, eps = d["N"], (1 if d["order"] == "B" else -1)
    want = {1: N + 1, -1: N + 1, 0: -(2 * N + 2 + eps)}
    flip = {k: -v for k, v in want.items()}
    if d["delta"] not in (want, flip) or N != (p - 3) // 4:
        bad += 1
        print("  extreme row mismatch", (p, b, x["r"]), d["delta"], want)
print(f"(4) Remark 2.7.3 on {seen} extreme rows: mismatches {bad}")


bad, seen = 0, 0
for p in range(5, 18, 2):
    for b in range(0, p - 3):
        for r in range(1, p + 1):
            seen += 1
            for s in range(5, p + 1):
                r1 = (r % p) + 1
                a0 = r + s - 5 if s <= 4 + b else r + s - 1
                a1 = r1 + s - 5 if s <= 4 + b else r1 + s - 1
                if T.mod(a1, p) != T.mod(a0 + 1, p):
                    bad += 1
print(f"(5) Lemma 2.7.1, tau_alpha: r -> r+1 on {seen} tuples, mismatches {bad}")


row = []
for r in range(1, 6):
    walk = T.traverse(7, 3, r)
    if walk is None:
        row.append((r, "disconnected"))
        continue
    e = sum(1 if t[0] == "L" else -1 for _, t, h in walk if t[0] != h[0])
    row.append((r, f"e={e:+d}"))
print("(6) orbit of D[7,2,3,1] under tau_alpha:",
      ", ".join(f"r={r}: {v}" for r, v in row))


f = dict(branch=0, values=0, top=0, band=0, iff_b=0)
for x in C.mix:
    d = scalars(x)
    p, b, S, k = x["p"], x["b"], d["S"], d["kappa"]
    g = T.gradings(p, b, x["r"])
    vals = set(g["A"].values())
    if S != 0 and d["cnt"] != 1:
        f["branch"] += 1
    if S == 0 and vals != {-1, 0, 1}:
        f["values"] += 1
    if S == 0 and d["top"] != 1:
        f["top"] += 1
    if S == 0 and k not in (0, 1):
        f["band"] += 1
    if (S == 0) != (b in (0, p - 4)):
        f["iff_b"] += 1
print("(7) dichotomy clauses (6)-(7) over 46 mixed tuples: "
      + ", ".join(f"{k} {v}" for k, v in f.items()))


import main2 as M
f8 = 0
want = [("coherent", None, 5, 1), ("mixed", -2, 2, 1),
        ("mixed", 0, 1, 2), ("mixed", 0, 1, 2)]
for (p, a, b, r), (kind, S, top, mult) in zip(M.CASES, want):
    d = M.data(p, b, r)
    got = ("coherent" if d["coh"] else "mixed", d["S"], d["top"], d["mult"])
```

```
    if got != (kind, S, top, mult):
        f8 += 1
        print("  figure case mismatch", (p, b, r), got, (kind, S, top, mult))
    if not d["coh"] and (d["S"] == 0) != (b in (0, p - 4)):
        f8 += 1
print(f"(8) Figures 20-21, four drawn cases: mismatches {f8}")


f9 = dict(steps=0, span=0, parity=0, toprank=0, fibered=0)
for x in C.mix:
    p, b, r = x["p"], x["b"], x["r"]
    d = scalars(x)
    walk = T.traverse(p, b, r)
    seq = [t[1] for _, t, h in walk]
    F = {seq[k]: seq[(k + 1) % p] for k in range(p)}
    allowed = set()
    for v in (1, 3, (r - 5) % p, (r - 1) % p):
        allowed |= {v % p, (-v) % p}
    if any((F[h] - h) % p not in allowed for h in seq):
        f9["steps"] += 1
    A = T.gradings(p, b, r)["A"]
    top, mult = max(A.values()), 0
    mult = sum(1 for v in A.values() if v == top)
    if max(A.values()) - min(A.values()) != abs(d["S"]) + 2:
        f9["span"] += 1
    if d["S"] % 2:
        f9["parity"] += 1
    if mult != (1 if d["S"] != 0 else d["N"] + 1):
        f9["toprank"] += 1
    if (mult == 1) != (d["S"] != 0 or d["N"] == 0):
        f9["fibered"] += 1
print("(9) Lemma 4.5.2 / Theorem 4.5.4 / Corollary 4.5.5 over 46 tuples: "
      + ", ".join(f"{k} {v}" for k, v in f9.items()))
```

B.5. The tabulation figure.

LISTING 5. `tab2.py`: Figure 19, every cell read from `chk26.py` and `tworainbow.py`.

```
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt

import tworainbow as T

plt.rcParams.update({"font.family": "serif", "mathtext.fontset": "cm",
                     "pdf.fonttype": 42, "savefig.dpi": 300})

INK, RULE, SOFT = "#000000", "#000000", "#F2F2F2"

HEAD = [r"$p,b,r$", r"type", r"$\ell$", r"$\kappa$", r"$N$", r"$m$",
        r"$S$", r"$A_{\mathrm{top}}$", r"$\mathrm{\#}A_{\mathrm{top}}$",
        r"span"]


def rows():
    import chk26 as C
    out = []
    for x in sorted(C.mix, key=lambda z: (z["p"], z["b"], z["r"])):
        if x["e"] != 1:
            continue
```

```python
        o, ell = x["order"], x["ell"]
        N = x["n"]["Q2"] if o == "A" else x["n"]["P2"]
        m = x["m"]["Q2"] if o == "A" else x["m"]["P2"]
        A = T.gradings(x["p"], x["b"], x["r"])["A"]
        top = max(A.values())
        S = ell * N - m
        assert S % 2 == 0
        assert top == abs(S) // 2 + 1
        assert top - min(A.values()) == abs(S) + 2
        out.append(dict(p=x["p"], b=x["b"], r=x["r"], order=o, ell=ell,
                        kappa=ell, N=N, m=m, S=S, top=top,
                        cnt=sum(1 for v in A.values() if v == top),
                        span=top - min(A.values())))
    return out


def cells(d):
    return [rf"${d['p']},{d['b']},{d['r']}$",
            rf"$\mathrm{{{d['order']}}}$",
            rf"${d['ell']}$", rf"${d['kappa']}$", rf"${d['N']}$",
            rf"${d['m']}$", rf"${d['S']}$", rf"${d['top']}$",
            rf"${d['cnt']}$", rf"${d['span']}$"]


def main(out="fig_tab2"):
    R = rows()
    n = len(R)
    w = np.array([2.05, 1.00, 0.82, 0.90, 0.78, 0.80, 0.88, 1.05, 1.10,
                  0.95])
    edge = np.concatenate([[0.0], np.cumsum(w)])
    mid = 0.5 * (edge[:-1] + edge[1:])
    W = edge[-1]
    H = n + 2.6

    fig, ax = plt.subplots(figsize=(0.685 * W, 0.148 * H), dpi=300)
    fs = 6.4

    for k in range(n):
        if k % 2:
            ax.add_patch(plt.Rectangle((0.0, k + 0.5), W, 1.0, color=SOFT,
                                       lw=0, zorder=0))
    for x, h in zip(mid, HEAD):
        ax.text(x, -0.25, h, fontsize=fs, ha="center", va="center",
                color=INK, zorder=3)
    for k, d in enumerate(R):
        for x, c in zip(mid, cells(d)):
            ax.text(x, k + 1.0, c, fontsize=fs, ha="center", va="center",
                    color=INK, zorder=3)

    for y, lw in ((-0.95, 0.9), (0.35, 0.55), (n + 0.55, 0.9)):
        ax.plot([0.0, W], [y, y], color=RULE, lw=lw, zorder=4,
                solid_capstyle="butt")

    ax.set_xlim(-0.10, W + 0.10)
    ax.set_ylim(n + 1.05, -1.45)
    ax.axis("off")
    for ext in ("png", "pdf"):
        fig.savefig(f"{out}.{ext}", bbox_inches="tight", pad_inches=0.05,
                    facecolor="white")
    plt.close(fig)
    print(f"wrote {out}: {n} rows, all identities asserted")
    return R
```

```python

if __name__ == "__main__":
    main()
```

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